General Relativity: Intrinsic Geometry and the Algebra of Tensors

Based on lectures by Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Notes reconstructed from the lecture and organized by LazyingArt LLC with Video2Book.

Chapter 1

Intrinsic Geometry and the Algebra of Tensors

Tensor notation is useful only when it does some of the thinking for us. Well-designed notation behaves like a box of mechanical pieces: an upper index fits a lower index, unlike objects refuse to be added, and a legal contraction leaves behind an object whose transformation law can be read from the remaining indices. The immediate purpose of this machinery is geometric. We want to distinguish genuine curvature from curvature that appears only because the coordinates have been bent.

1.1 Curvature Seen from Inside

Begin with an ordinary sheet of paper. It may be rolled into a cylinder without stretching or tearing. To an observer in the surrounding room, the sheet is plainly bent; nevertheless, distances measured within the sheet are unchanged. Its extrinsic curvature has changed, while its intrinsic geometry has not.

Imagine now a small creature confined to the sheet. It cannot look into the ambient room. It can measure lengths, compare neighboring directions, construct triangles, and transport measuring rods from place to place. Those internal experiments are all that geometry should require. The corresponding question in gravitation is whether a field represents true spacetime curvature or merely a curvilinear description of flat spacetime.

One way to make this intrinsic viewpoint concrete is to replace the surface by a network of points and links. Join neighboring points, record the length of every link, and allow the joints to hinge. An equilateral triangular lattice lies flat. Bending the hinges does not alter the encoded geometry because the lattice can always be returned to the plane without changing a length.

Now lengthen the six links meeting at one vertex while holding the surrounding links fixed. The central vertex can no longer remain in the plane without violating the prescribed lengths. It must rise into a bulge. No view from outside was needed to diagnose this curvature: the incompatible neighboring distances already contained it.

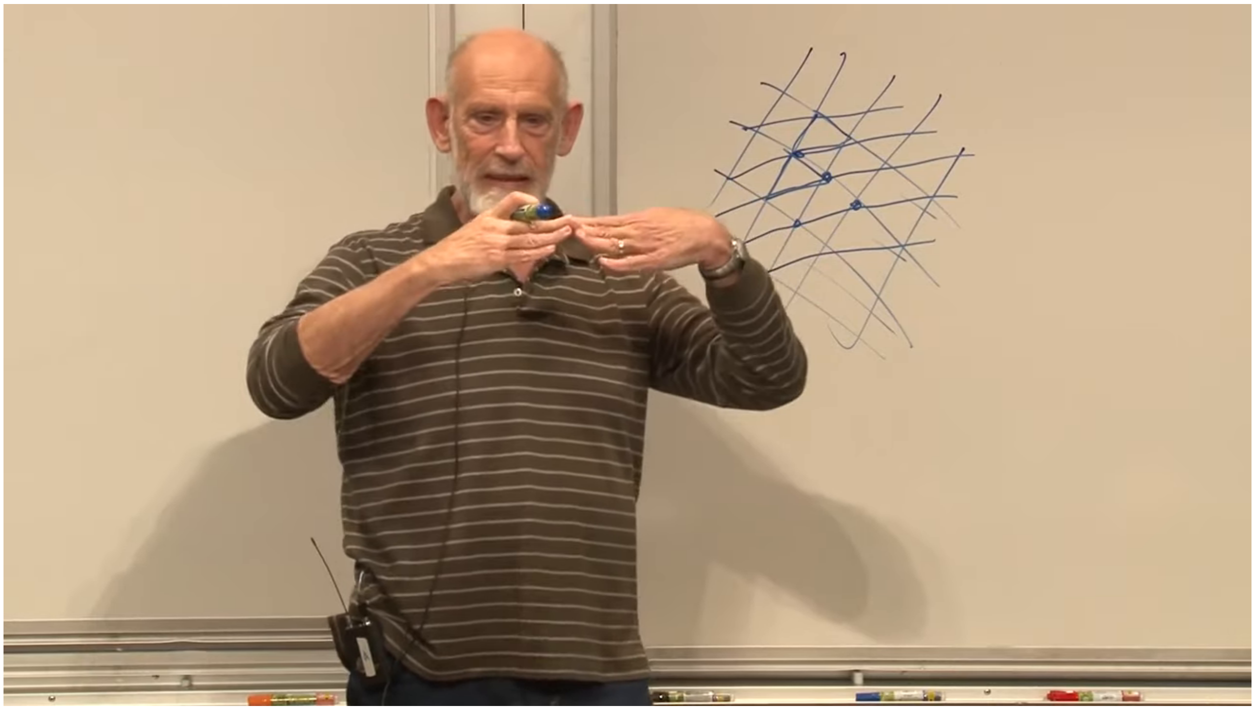

Figure 1.1: The triangular-link model of intrinsic geometry. Altering the links around one vertex forces a bulge that cannot be flattened without changing lengths.

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Question from the audience. What, then, is the precise definition of a flat space?

Response. The operational answer is that its intrinsic distance relations can be realized without distortion in a Euclidean layout. In the language developed below, a Riemannian space is flat when one can find coordinates, throughout the region of interest, in which the metric is the constant Kronecker delta. At a single point this can always be done locally; doing it over a region is the nontrivial condition. The curvature tensor will eventually state that condition without reference to a special coordinate choice. [a]

[a]Lecture timestamp: 00:09:15.

This answer uses the metric before we have earned it. We therefore turn temporarily from curvature to tensor analysis. The detour is not optional: the metric must be recognized as a tensor before its coordinate dependence can be separated from genuine geometry.

1.2 Fields and Coordinate Descriptions

A tensor in general relativity is usually a tensor *field*: at every point there is a tensor, and its components vary from point to point. A scalar field is the simplest example. Temperature on a surface may be represented by a number $S(P)$ at each point P. Two people may label the point by different coordinates, but they must agree on the temperature there.

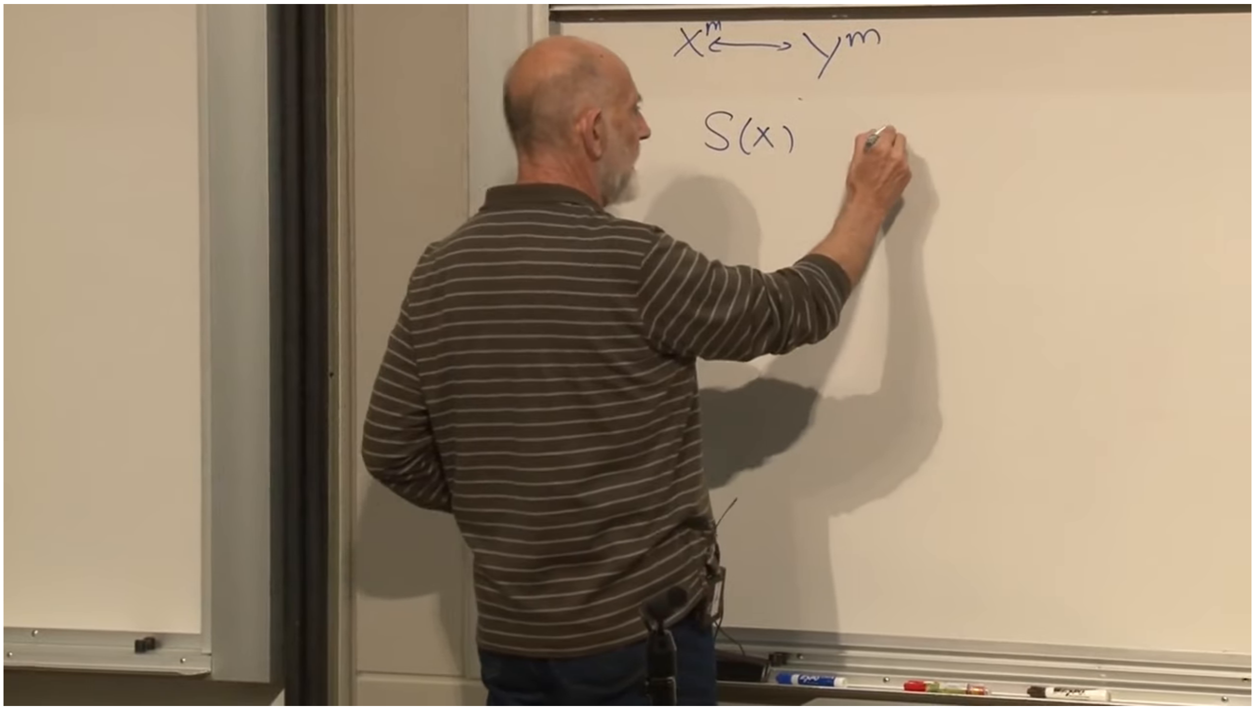

Figure 1.2: A scalar field together with two coordinate labels for the same points.
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Let x^m and y^m be two smooth coordinate systems. Each is a map from points to ordered lists of numbers, and in an overlapping coordinate patch the two lists are related by mutually inverse functions,

$$y^m = y^m(x^1, \ldots, x^d), \qquad x^m = x^m(y^1, \ldots, y^d). \tag{1.1}$$

Smoothness allows differentiation, and invertibility ensures that no point is lost or identified with another merely by changing coordinates.

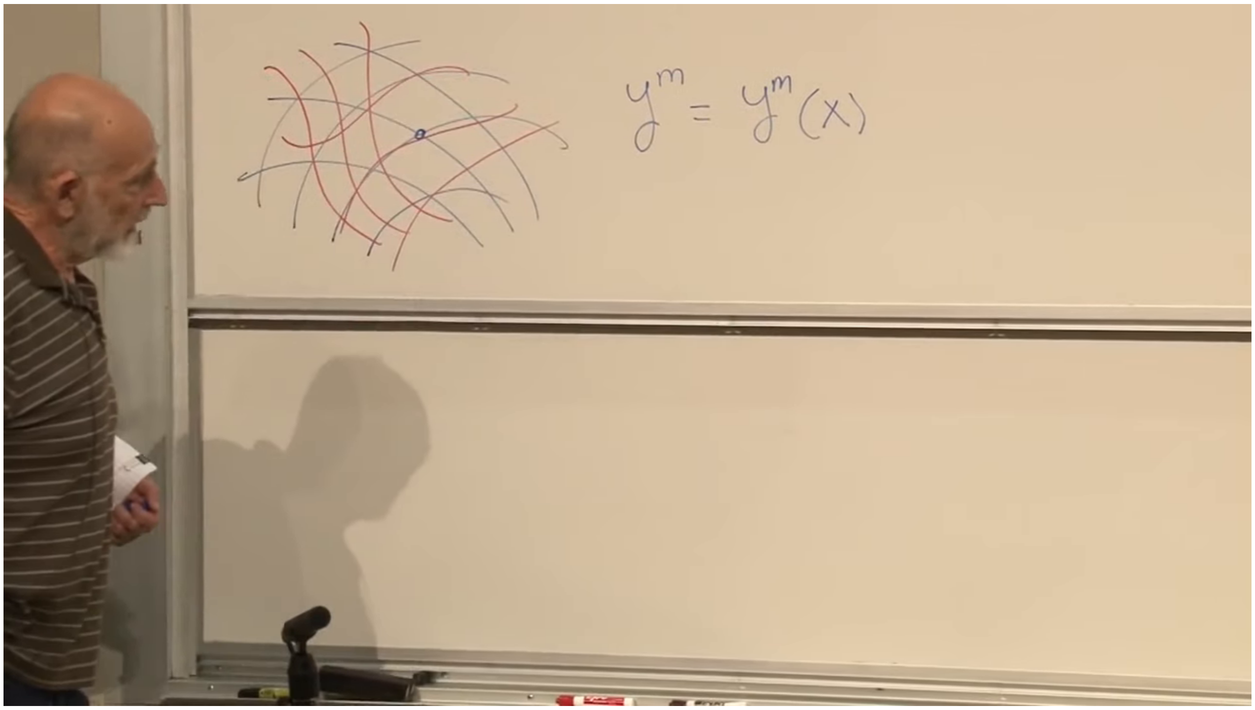

Figure 1.3: Two smooth, invertible coordinate maps describing the same geometric region.
Lecture frame: 00:13:10.

For a scalar field the transformation law is simply

$$S'(y) = S(x). \tag{1.2}$$

The functions on the two sides may look different because their arguments differ, but at corresponding points their values agree. This is the prototype of an invariant statement.

1.3 Oblique Coordinates and Two Kinds of Components

Before defining general tensors by transformation laws, it is useful to see why upper and lower vector indices cannot be treated as decorative variants of one another.

Draw a coordinate grid whose axes are neither perpendicular nor normalized. At a point, increasing x^m by one coordinate unit produces a basis vector $\mathbf{e}_m$. In coordinate language,

$$\mathbf{e}_m = \frac{\partial \mathbf{r}}{\partial x^m}. \tag{1.3}$$

The vectors $\mathbf{e}_m$ need not have unit length and need not be orthogonal.

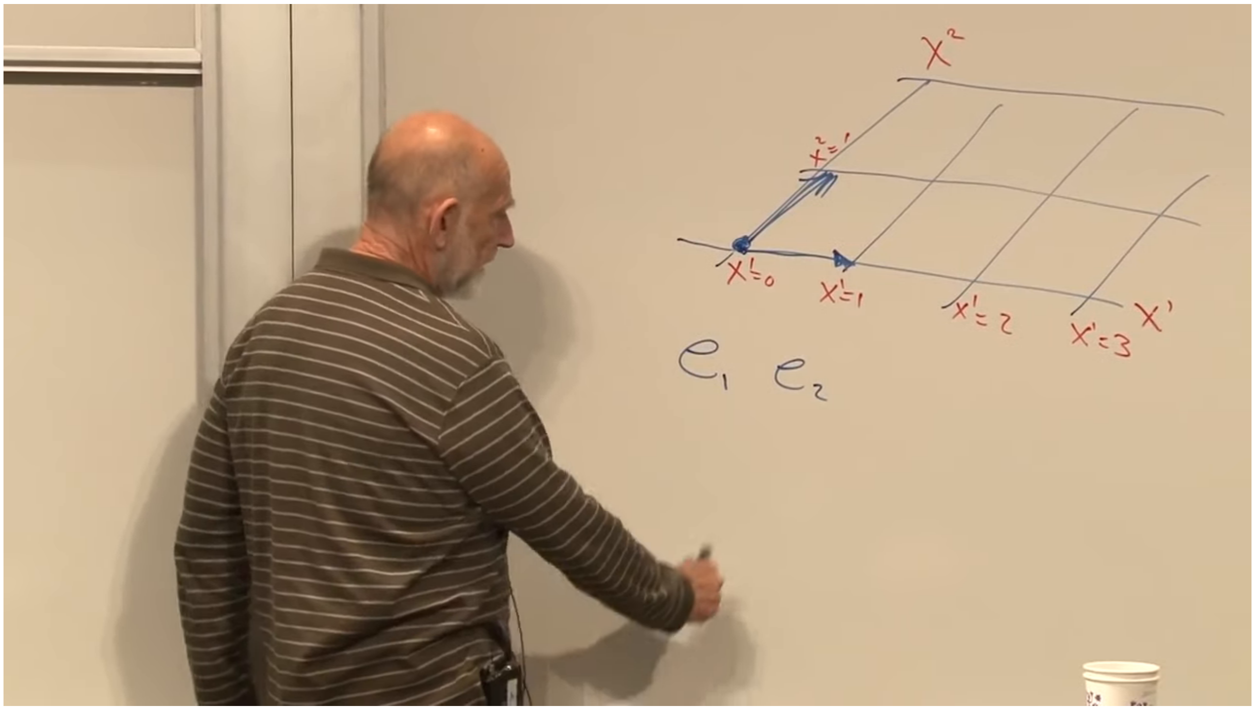

Figure 1.4: An oblique coordinate grid and its coordinate basis vectors.
Lecture frame: 00:17:40.

Every vector **V** can be expanded in this basis:

$$\mathbf{V} = V^m \mathbf{e}_m. \tag{1.4}$$

The numbers V^m are the *contravariant components*. They are the coefficients needed to assemble the geometric vector from the basis vectors.

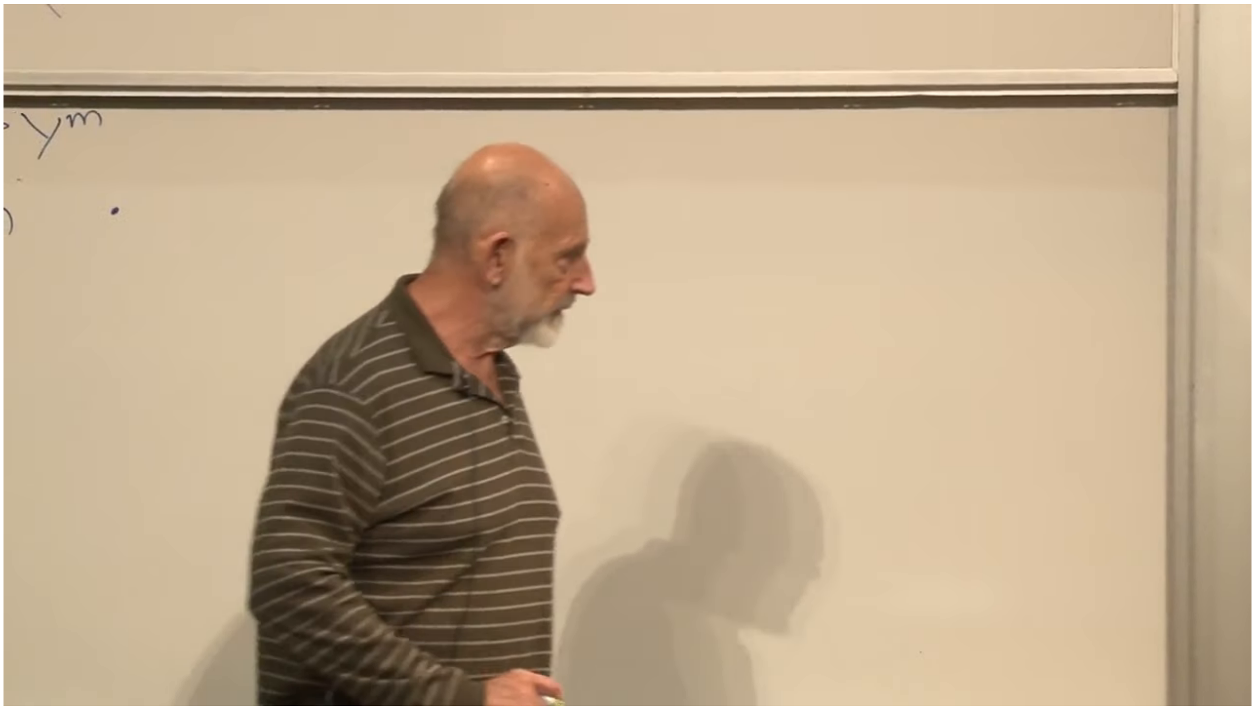

Figure 1.5: The expansion of a vector in a nonorthogonal coordinate basis.
Lecture frame: 00:19:20.

There is another natural set of numbers:

$$V_m = \mathbf{V} \cdot \mathbf{e}_m. \tag{1.5}$$

These are the *covariant components*. In an oblique basis they are not simply the same coefficients with the index moved. Substituting (1.4) gives

$$V_m = V^n(\mathbf{e}_n \cdot \mathbf{e}_m). \tag{1.6}$$

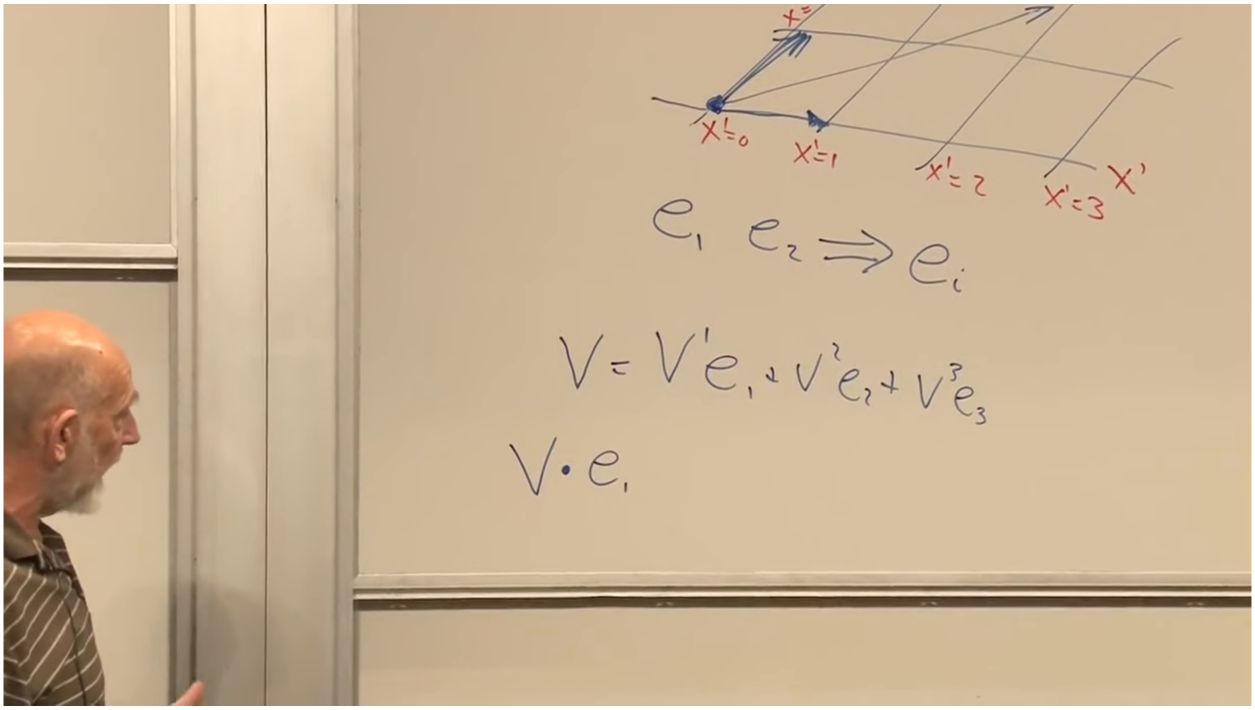

Figure 1.6: Covariant components obtained by taking scalar products with the coordinate basis.
Lecture frame: 00:21:10.

The scalar products of the basis vectors form the Gram matrix

$$g_{mn} = \mathbf{e}_m \cdot \mathbf{e}_n. \tag{1.7}$$

Consequently,

$$V_m = g_{mn} V^n. \tag{1.8}$$

This is the first appearance of the metric. It converts expansion coefficients into projections because it records both the lengths and mutual angles of the coordinate basis.

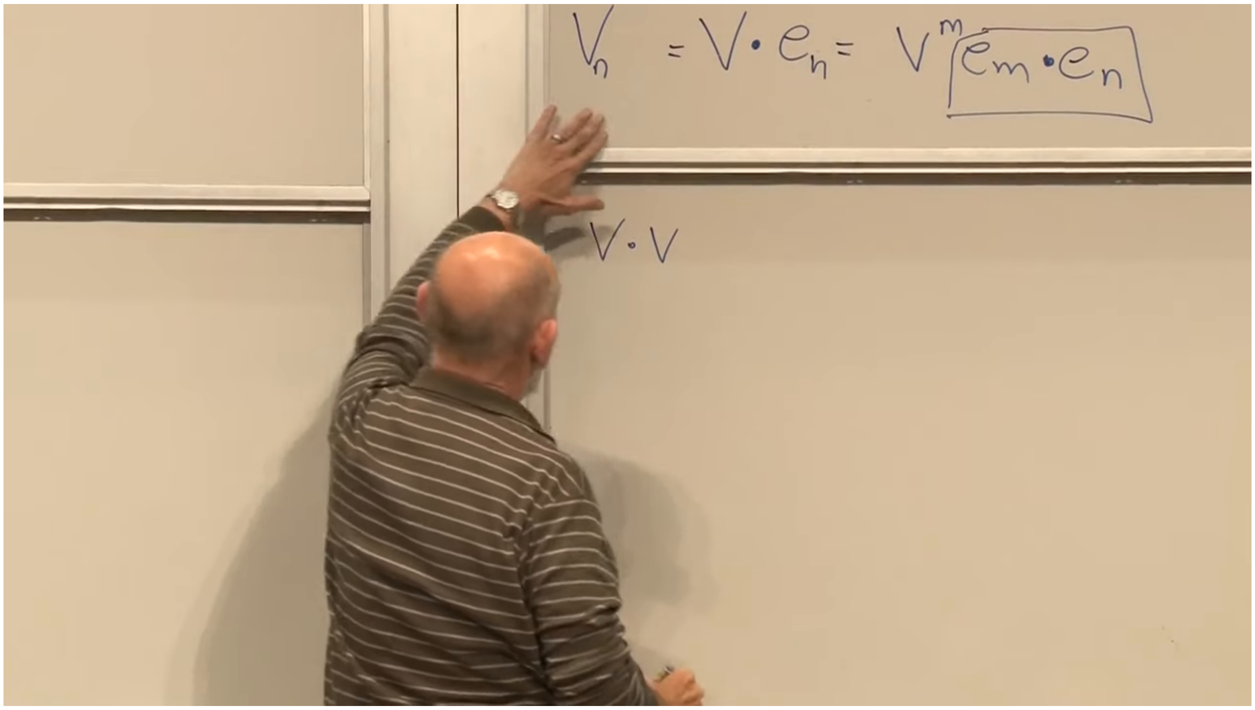

Figure 1.7: The metric as the Gram matrix of the basis vectors.
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The same matrix computes length:

$$\begin{aligned}\mathbf{V}\cdot\mathbf{V} &= (V^m\mathbf{e}_m)\cdot(V^n\mathbf{e}_n)\\ &= g_{mn}V^mV^n.\end{aligned} \tag{1.9}$$

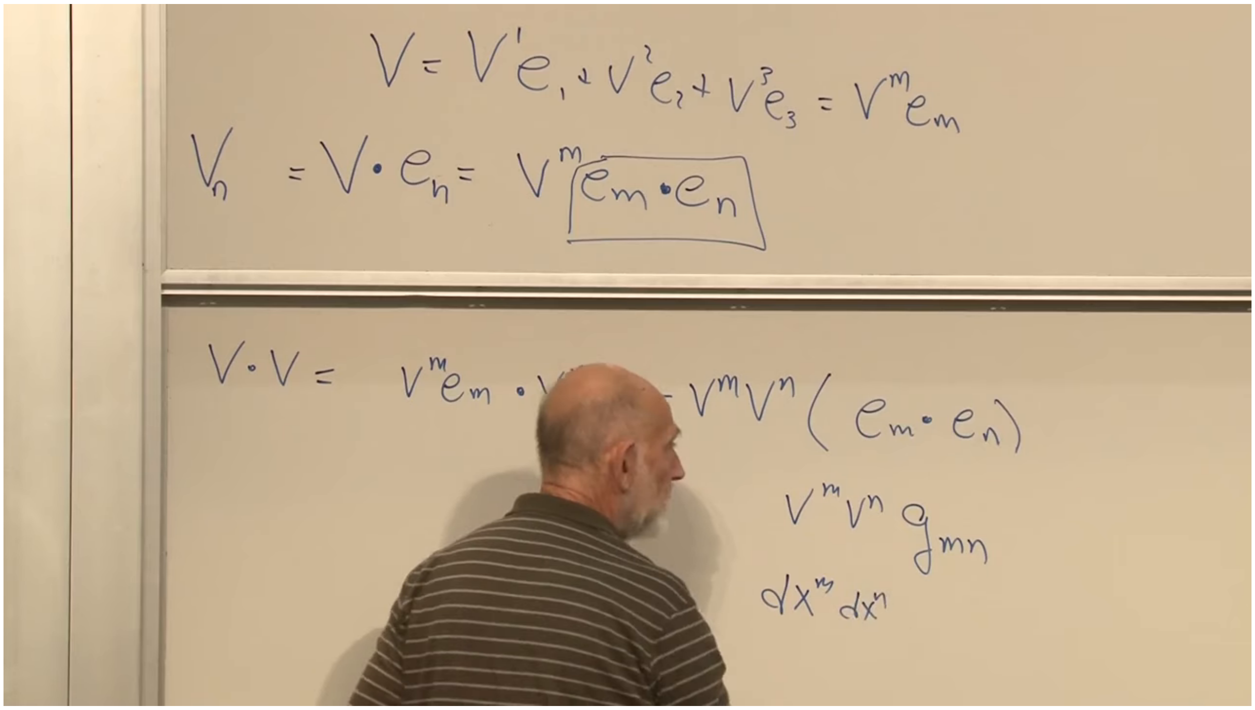

Figure 1.8: The vector norm written in terms of the metric and contravariant components.
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Question from the audience. Why introduce this geometric construction before the formal transformation laws?

Response. It gives the indices a concrete meaning. Upper components are expansion coefficients, lower components are linear measurements of the vector, and the metric connects the two. The formal definition will then tell us that these roles survive arbitrary smooth coordinate changes. [a]

[a]Lecture timestamp: 00:26:07.

Question from the audience. When do the covariant and contravariant components have the same numerical values?

Response. In an orthonormal Euclidean basis, $g_{mn} = \delta_{mn}$, so $V_m = V^m$ component by component. Mere orthogonality is not enough: if the basis vectors have nonunit lengths, the diagonal metric still supplies scale factors. [a]

[a]Lecture timestamp: 00:26:19.

Polar coordinates provide the clean warning. Radial and angular coordinate lines meet orthogonally, yet a fixed change $d\theta$ corresponds to physical distance $r\,d\theta$. In the plane,

$$ds^2 = dr^2 + r^2 d\theta^2, \tag{1.10}$$

so the basis is orthogonal but not Cartesian.

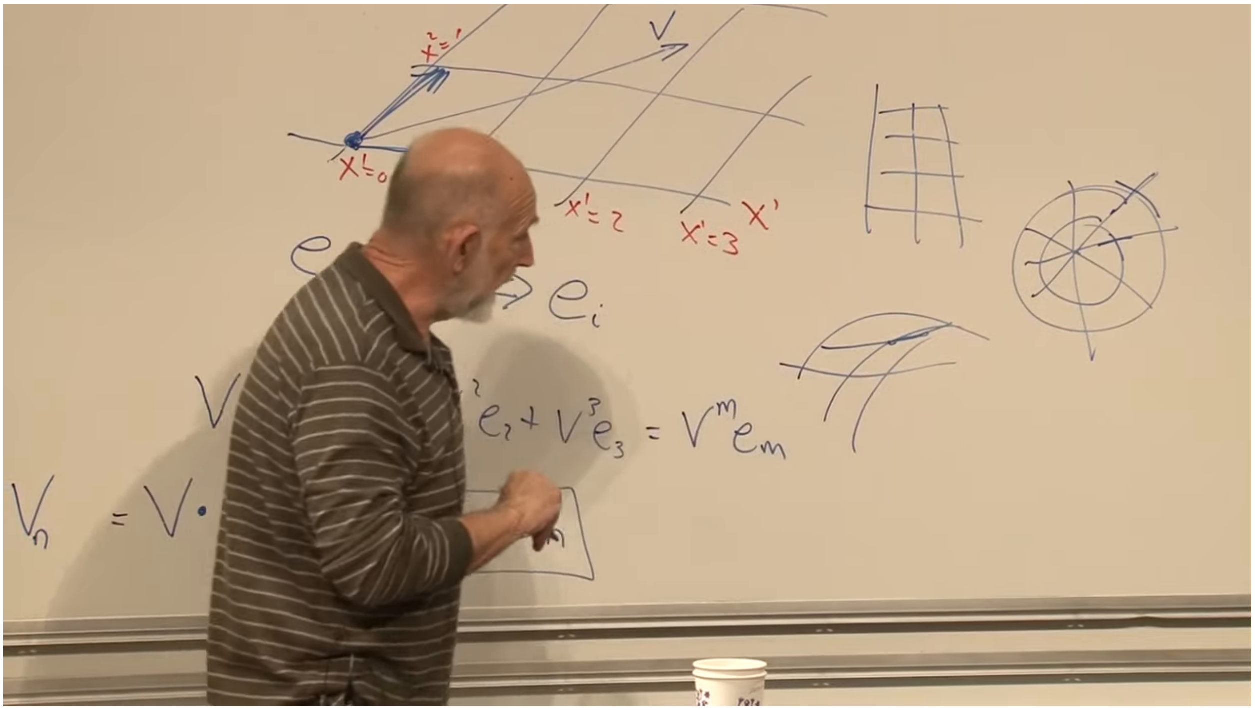

Figure 1.9: Polar coordinates: orthogonal coordinate directions with an angular scale factor that grows with radius.

Lecture frame: 00:28:10.

Question from the audience. Are covariant vectors genuinely different vectors, or only a second way of listing the same vector?

Response. Before a metric is supplied, a vector and a covector are different kinds of linear object. A vector belongs to the tangent space; a covector is a linear functional on that space. A nondegenerate metric gives a one-to-one correspondence between them, represented by $V_m = g_{mn}V^n$, but the correspondence should not be mistaken for identity. [a]

[a] Lecture timestamp: 00:29:10.

1.4 Transformation Laws Define the Tensor

The safest formal definition is operational: an array is a tensor if its components transform by the appropriate Jacobian factors. Differentiating (1.1) gives

$$dy^m = \frac{\partial y^m}{\partial x^n}\, dx^n. \tag{1.11}$$

Any contravariant vector obeys the same rule,

$$V'^m(y) = \frac{\partial y^m}{\partial x^n} V^n(x). \tag{1.12}$$

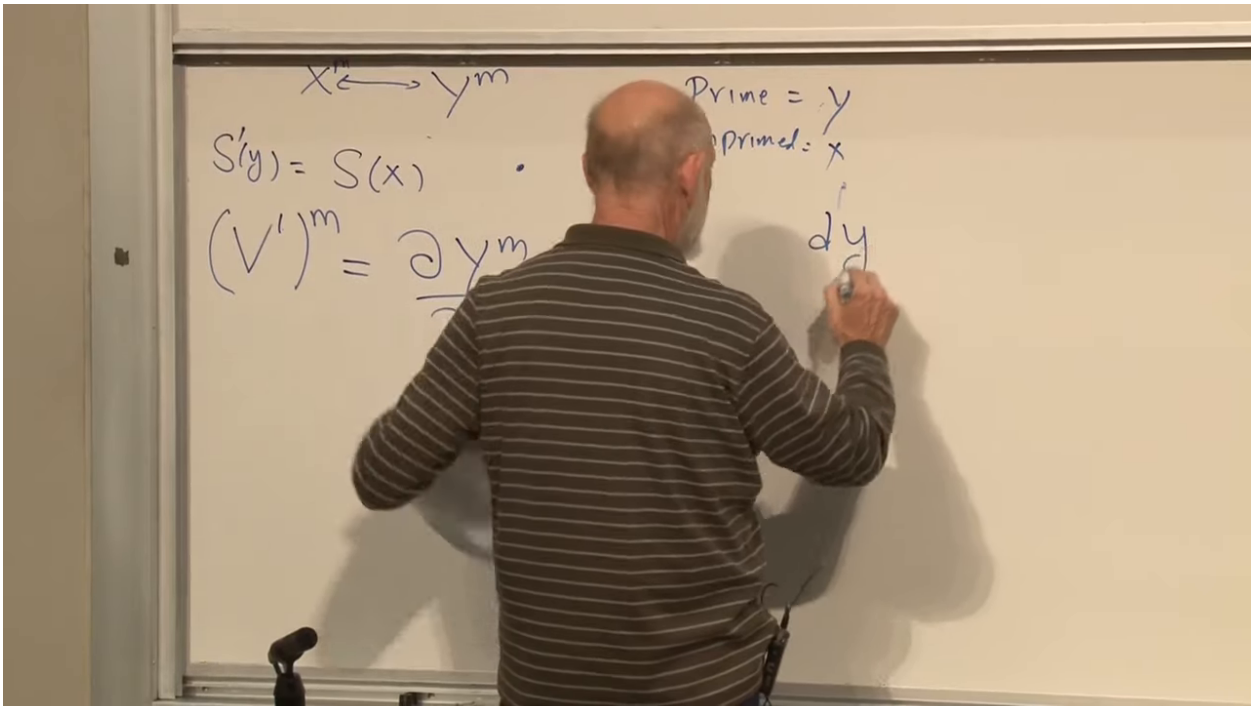

Figure 1.10: The contravariant transformation law, modeled on the transformation of coordinate differentials.

Lecture frame: 00:32:50.

A covector transforms with the inverse Jacobian:

$$W'_m(y) = \frac{\partial x^n}{\partial y^m} W_n(x). \tag{1.13}$$

The gradient of a scalar is the standard example. Applying the chain rule to (1.2),

$$\frac{\partial S}{\partial y^m} = \frac{\partial x^n}{\partial y^m} \frac{\partial S}{\partial x^n}. \tag{1.14}$$

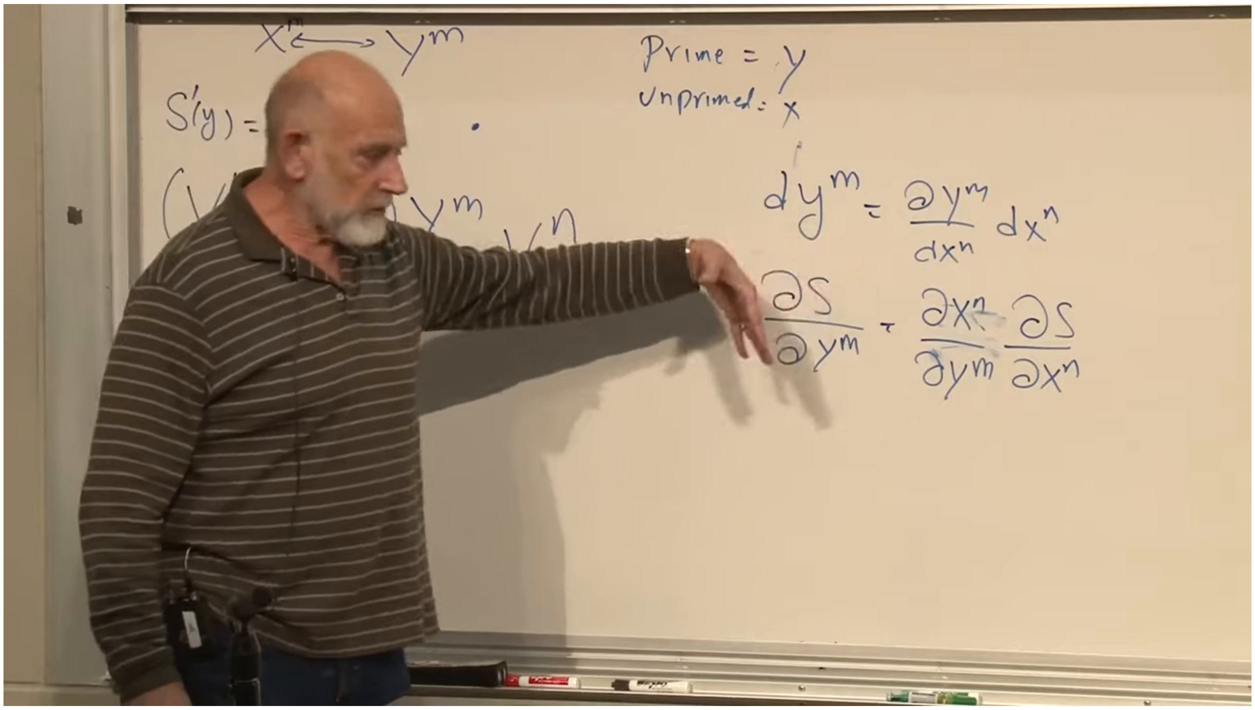

Figure 1.11: A scalar gradient transforming as a covector by the chain rule.

Lecture frame: 00:34:20.

Question from the audience. Do upper and lower components necessarily have the same physical units?

Response. No. Coordinate components inherit the units and scale conventions of the coordinates and their basis vectors. If one coordinate is a length and another an angle, their components need not even share dimensions. The complete contraction is the object whose physical dimensions are coordinate independent. [a]

[a] Lecture timestamp: 00:35:41.

Question from the audience. Why are the derivatives partial derivatives rather than ordinary derivatives?

Response. A field depends on several coordinates. The Jacobian asks how one coordinate function changes when one argument varies and the others are held fixed, so its entries are partial derivatives. [a]

[a] Lecture timestamp: 00:36:45.

The pattern now extends mechanically. Every upper primed index brings one factor of $\partial y/\partial x$; every lower primed index brings one factor of $\partial x/\partial y$. A mixed tensor transforms as

$$T'^m{}_n = \frac{\partial y^m}{\partial x^p}\frac{\partial x^q}{\partial y^n} T^p{}_q, \tag{1.15}$$

and a rank-two covariant tensor transforms as

$$W'_{mn} = \frac{\partial x^p}{\partial y^m}\frac{\partial x^q}{\partial y^n} W_{pq}. \tag{1.16}$$

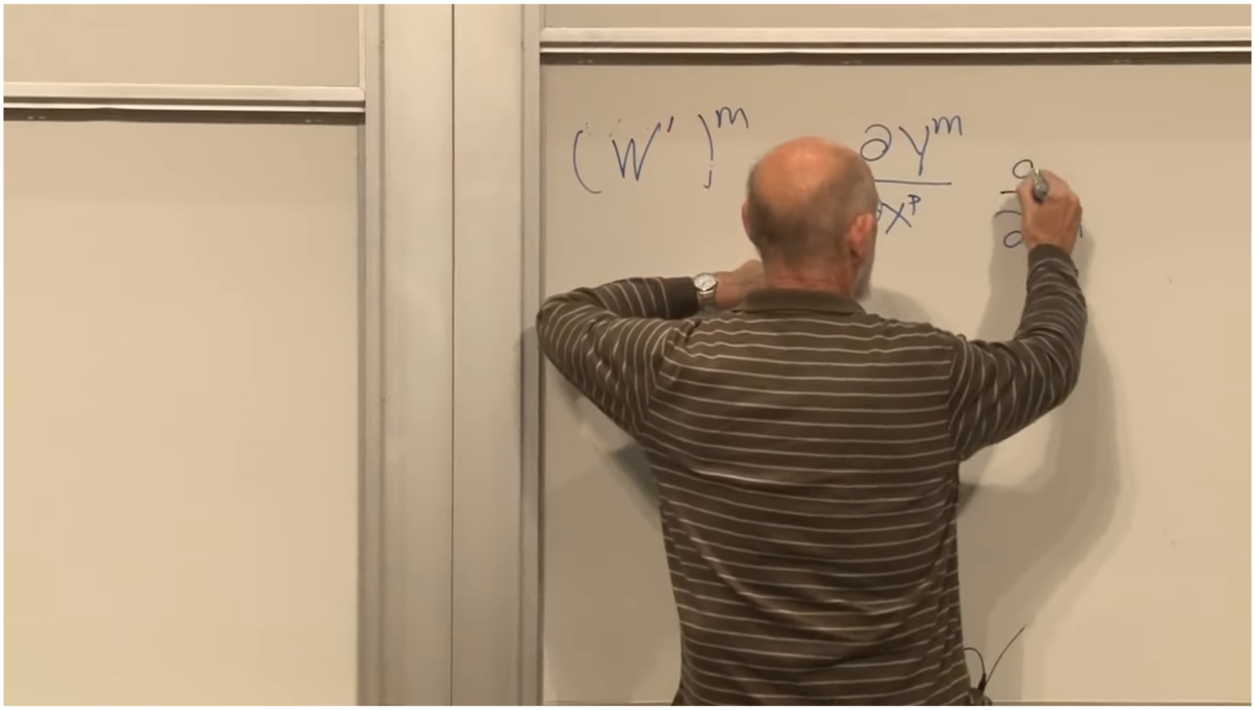

Figure 1.12: The transformation rule for a tensor with one upper and one lower index.

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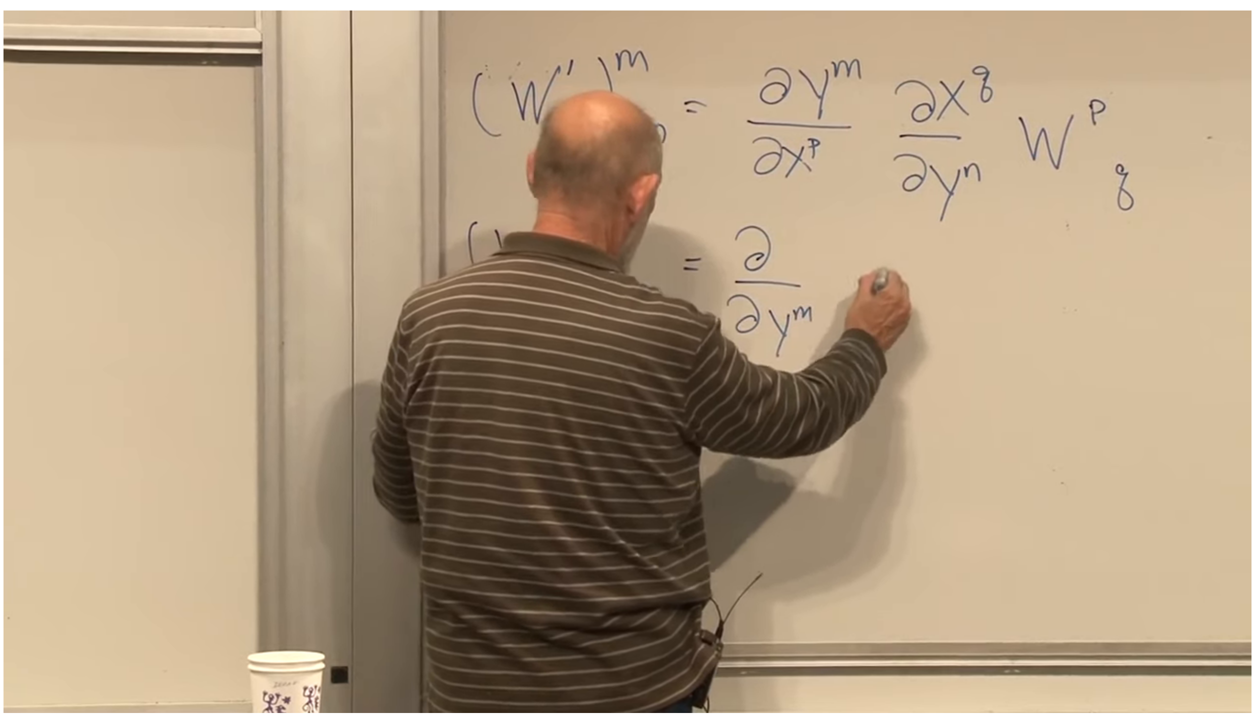

Figure 1.13: A rank-two covariant transformation law, with one inverse Jacobian for each lower index.

Lecture frame: 00:42:15.

Repeated indices are summed unless stated otherwise. This convention removes summation signs while preserving all the information needed to reconstruct them. A repeated dummy index must occur exactly twice in a term, once upstairs and once downstairs; a free index occurs once in every term of an equation.

Question from the audience. Who introduced this summation notation?

Response. It is conventionally credited to Einstein, which is why it is called the Einstein summation convention. Its value is not historical ornament: the index pattern makes legal tensor operations visible at a glance. [a]

[a] Lecture timestamp: 00:43:21.

1.5 Tensor Equations and Coordinate-Independent Truth

Components vary with coordinates, but a tensor equation remains true in every coordinate system. If two tensors of the same type satisfy $W = T$, then their difference is the zero tensor:

$$W^{m\cdots}{}_{\cdots n} - T^{m\cdots}{}_{\cdots n} = 0. \tag{1.17}$$

The transformation law is linear, so zero components in one system produce zero components in every system. The matching free indices on the two sides are indispensable. An equation between objects of different tensor type has no coordinate-independent meaning.

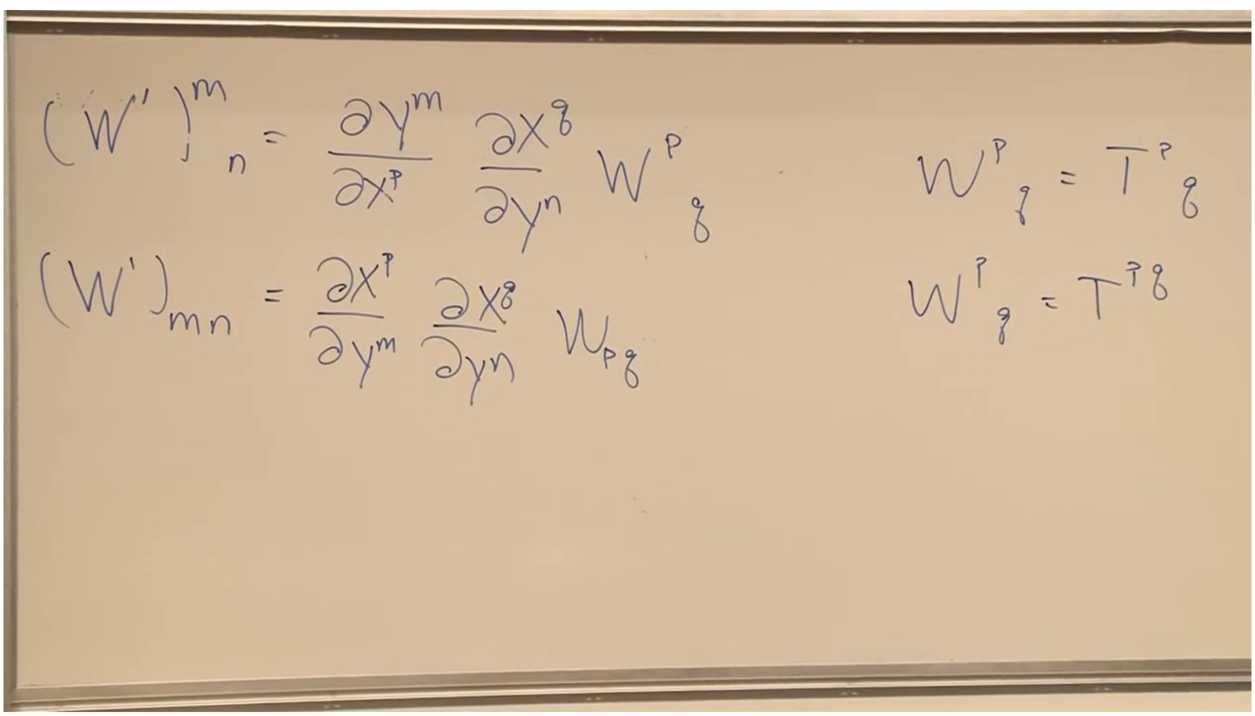

Figure 1.14: Matching index structure turns equality of components into a tensor equation valid in every coordinate system.

Lecture frame: 00:46:35.

Question from the audience. What exactly is invariant if the vector components are changing?

Response. The geometric tensor and the truth of a tensor equation are invariant; the coordinate list is not. A scalar is the special case with no indices, so its numerical value at a point is the same in every coordinate system. [a]

[a]Lecture timestamp: 00:47:19.

Question from the audience. Is vector magnitude the invariant analogue, and does zero magnitude mean the vector is zero?

Response. Once a metric is given, the contraction $g_{mn}V^mV^n$ is a scalar and therefore an invariant squared norm. In positive-definite geometry it vanishes only for the zero vector. In Lorentzian spacetime a nonzero null vector can have zero norm, so the two statements must be kept separate. [a]

[a]Lecture timestamp: 00:48:04.

1.6 The Permitted Algebra of Tensors

The board collects the principal operations that will be needed: multiply by a scalar, add or subtract tensors of the same type, form tensor products, contract an upper with a lower index, and eventually differentiate covariantly.

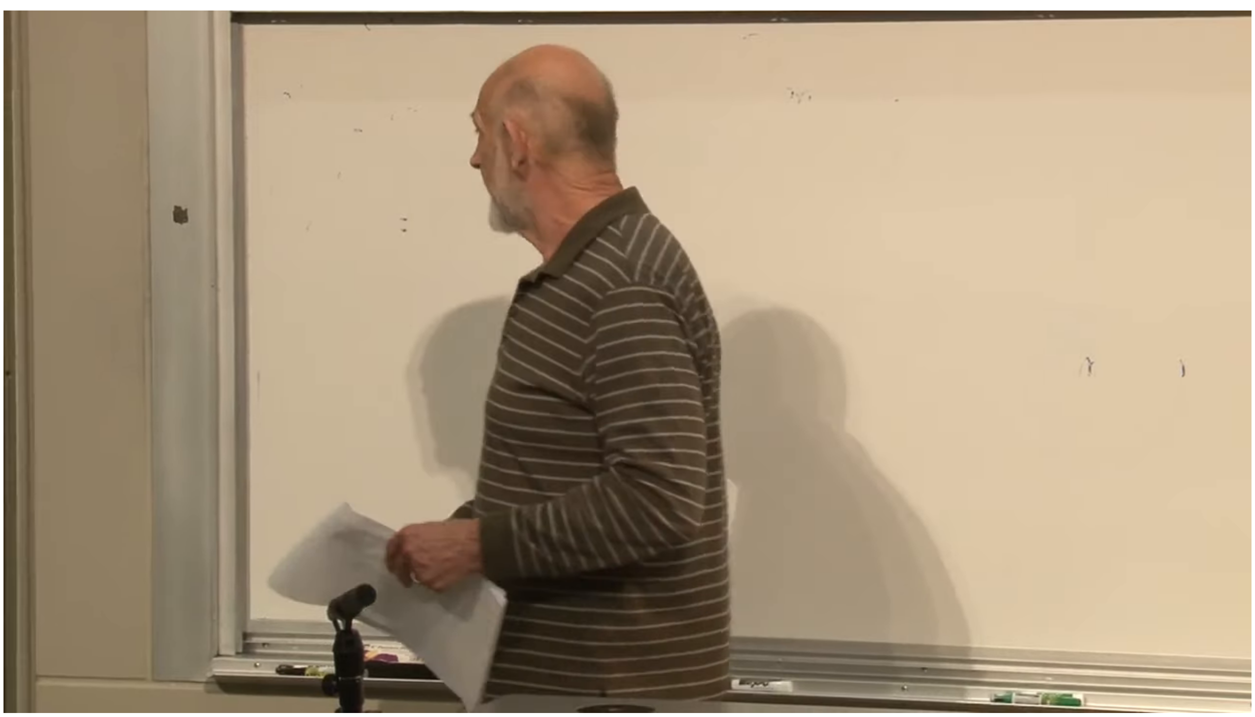

Figure 1.15: The board inventory of tensor operations: addition, multiplication, contraction, and differentiation.

Lecture frame: 00:50:40.

Question from the audience. Differentiation with respect to what?

Response. With respect to position, hence with respect to the coordinates. Ordinary component derivatives do not generally transform as tensors because the Jacobians themselves vary. Covariant differentiation repairs this, but its construction is postponed until the geometric preliminaries are in place. [a]

[a] Lecture timestamp: 00:51:49.

Scalar multiplication is immediate. Addition and subtraction require matching tensor type:

$$C^m{}_n = A^m{}_n + B^m{}_n, \qquad D^m{}_n = A^m{}_n - B^m{}_n. \tag{1.18}$$

One cannot add a vector to a covector or a rank-two tensor to a scalar, just as one cannot add a velocity to an area.

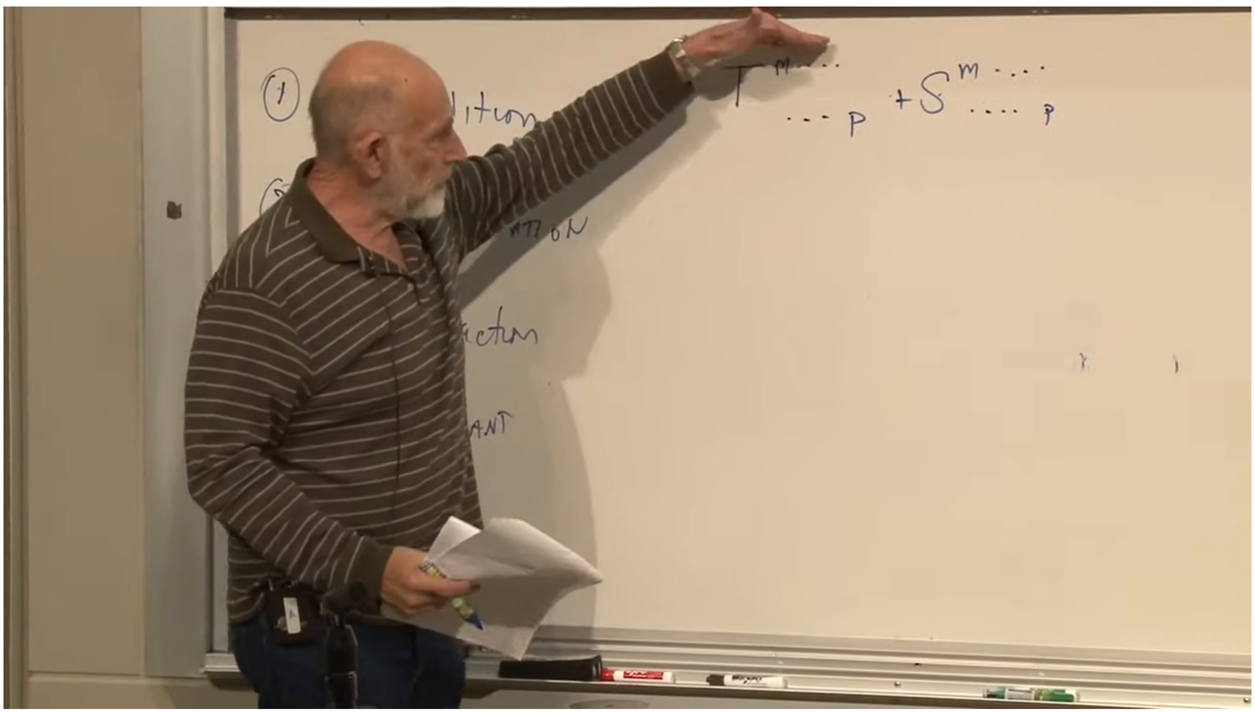

Figure 1.16: Addition is defined componentwise only for tensors with the same free-index pattern.
Lecture frame: 00:53:00.

The tensor product combines all indices of its factors. For example,

$$T^{mn} = V^m W^n, \qquad S^m{}_n = V^m U_n. \tag{1.19}$$

In four dimensions T^{mn} has sixteen components. No index has been summed, so this is an outer product rather than a dot product.

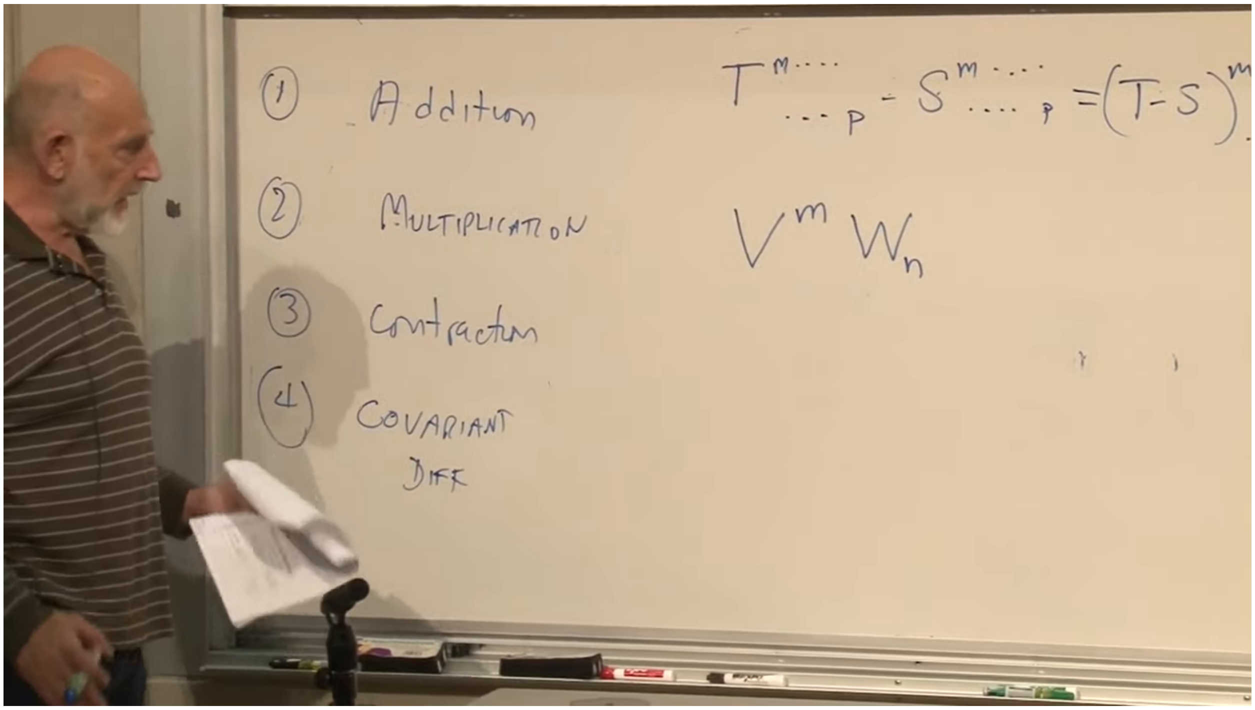

Figure 1.17: A tensor product retaining one free index from each vector factor.
Lecture frame: 00:55:15.

Question from the audience. Is multiplication of two vectors here the dot product?

Response. No. A dot product contracts an index and produces a scalar. The tensor product leaves both indices free and therefore produces a rank-two tensor. In four dimensions that is sixteen components rather than one. [a]

[a]Lecture timestamp: 00:56:36.

Question from the audience. Since the component numbers commute, does exchanging the two vector factors make no difference?

Response. For ordinary scalar components, $V^m W^n = W^n V^m$. But tensor slots are ordered. Interchanging the factors and then naming the first slot m and the second n produces the transposed array; T^{mn} need not equal T^{nm}. [a]

[a]Lecture timestamp: 00:58:52.

1.7 Contraction and the Inverse-Jacobian Lemma

The key to contraction is the chain rule. Because the coordinate maps are inverse,

$$\frac{\partial x^b}{\partial y^m}\frac{\partial y^m}{\partial x^a} = \frac{\partial x^b}{\partial x^a} = \delta^b{}_a. \tag{1.20}$$

The two Jacobian matrices are inverses, and their product is the Kronecker delta.

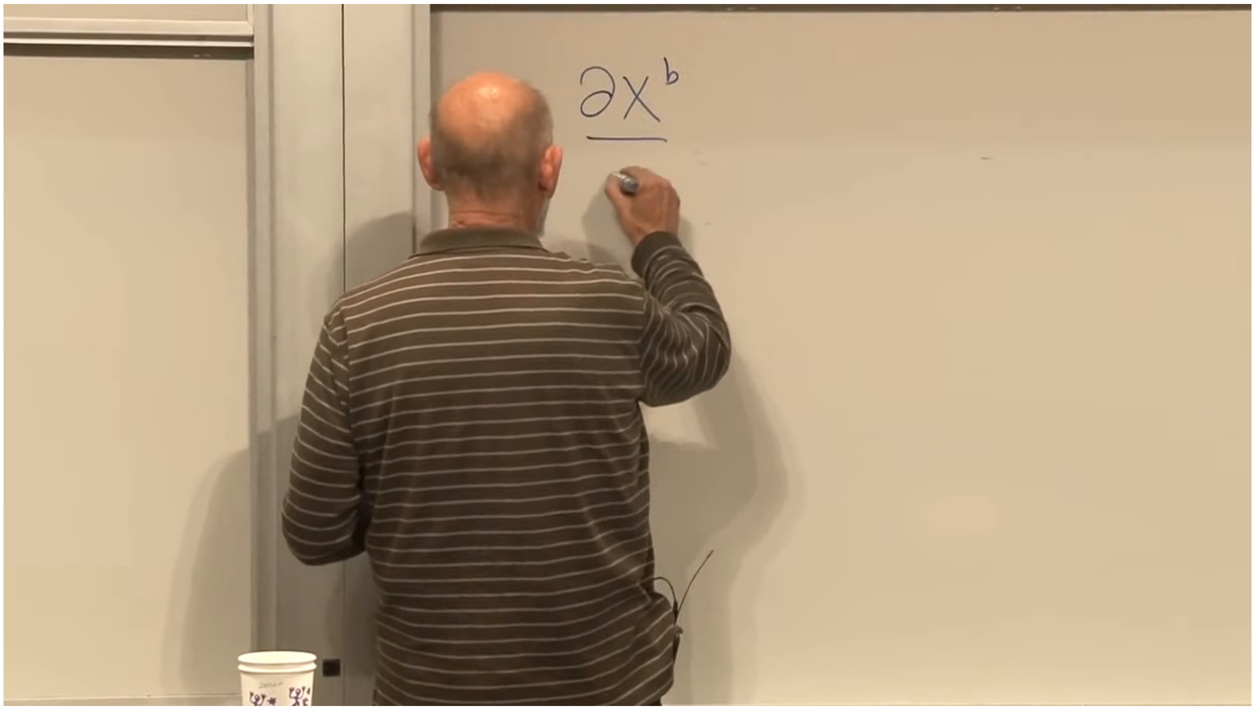

Figure 1.18: The inverse-Jacobian identity prepared as the lemma behind tensor contraction.
Lecture frame: 01:02:15.

Question from the audience. What kind of object is the inverse Jacobian?

Response. It is the derivative matrix of the inverse coordinate map. Its entries are $\partial x^b/\partial y^m$, and matrix multiplication with $\partial y^m/\partial x^a$ gives the identity by the multivariable chain rule. [a]

[a] Lecture timestamp: 01:02:37.

A quick proof makes the notation transparent. For any scalar function f,

$$\frac{\partial f}{\partial y^m}\frac{\partial y^m}{\partial x^a} = \frac{\partial f}{\partial x^a}. \tag{1.21}$$

Choosing $f = x^b$ gives

$$\frac{\partial x^b}{\partial y^m}\frac{\partial y^m}{\partial x^a} = \frac{\partial x^b}{\partial x^a} = \delta^b{}_a. \tag{1.22}$$

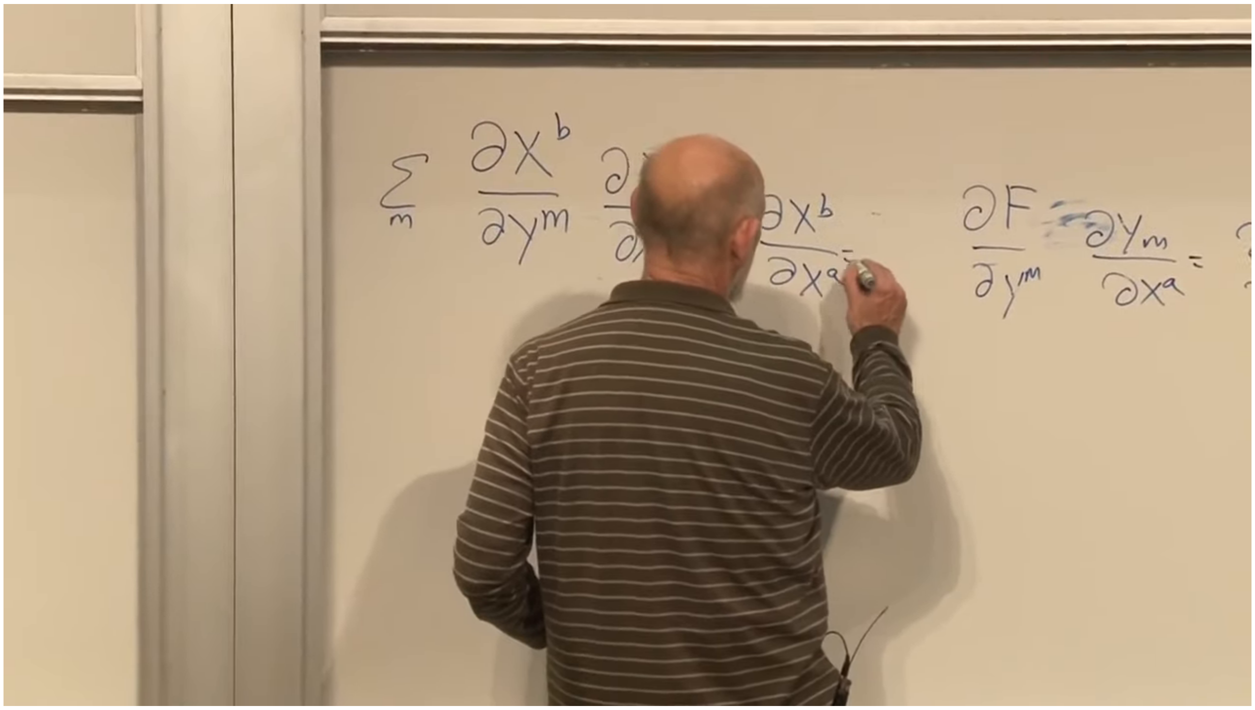

Figure 1.19: The chain rule collapsing the inverse Jacobians to the Kronecker delta.
Lecture frame: 01:04:55.

Question from the audience. Is $\delta^b{}_a$ the identity only in Cartesian coordinates?

Response. No. The mixed tensor $\delta^b{}_a$ represents the identity map in every coordinate system. A Euclidean metric written as δ_{ab} is special to orthonormal Cartesian coordinates, but the mixed-index identity should not be confused with that metric. [a]

[a] Lecture timestamp: 01:04:58.

Now take a mixed tensor $T^m{}_n$. Its transformed trace is

$$
\begin{aligned}
T'^m{}_m &= \frac{\partial y^m}{\partial x^a}\frac{\partial x^b}{\partial y^m}T^a{}_b \\
&= \delta^b{}_a T^a{}_b = T^a{}_a.
\end{aligned}
\tag{1.23}
$$

The contracted quantity has no free index and is therefore a scalar.

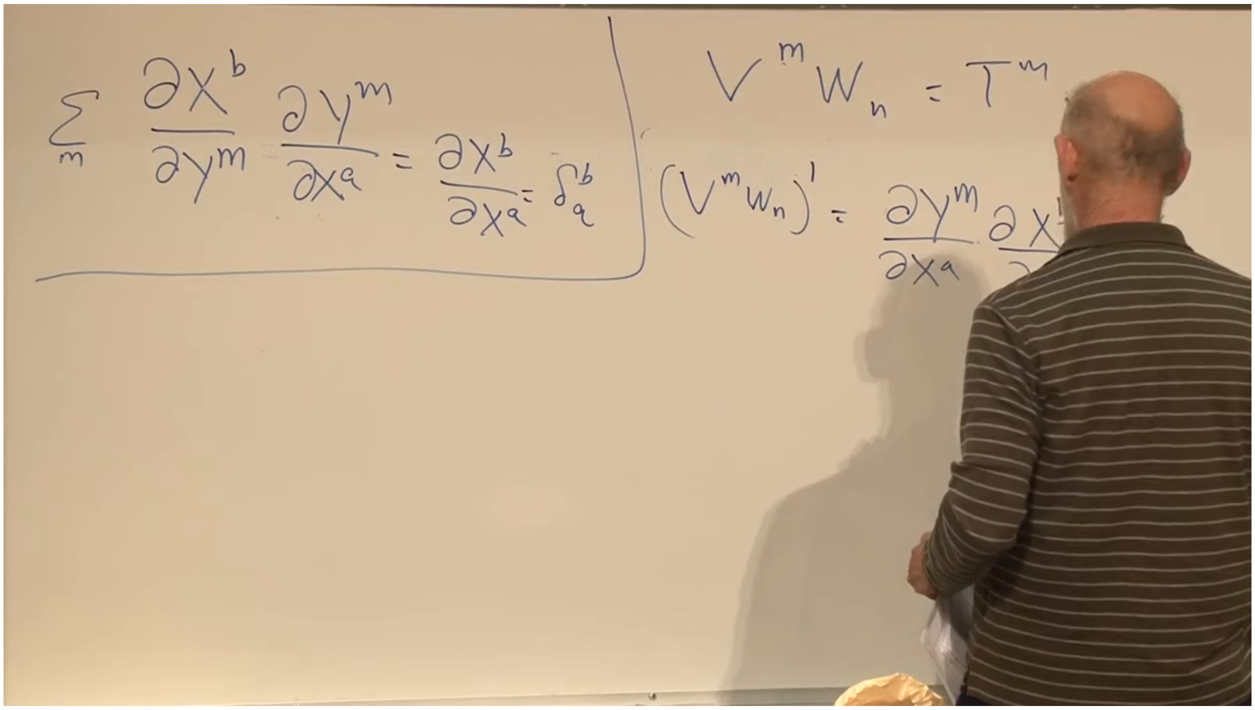

Figure 1.20: Transformation of a mixed tensor before the upper and lower indices are contracted.
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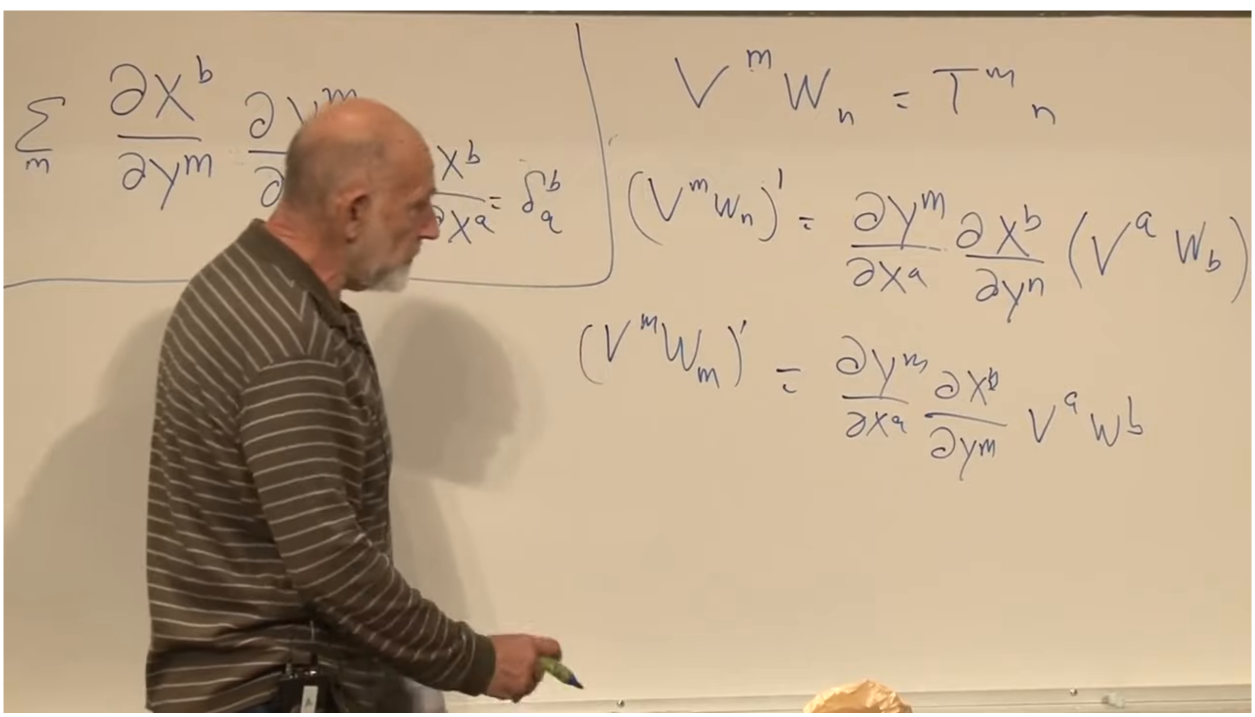

Figure 1.21: The Jacobians cancel under contraction, leaving the same scalar trace in either coordinate system.
Lecture frame: 01:10:30.

Question from the audience. After a contraction, does the result still carry indices?

Response. Only the contracted pair disappears. Contracting the sole upper and lower indices of $T^m{}_n$ leaves a scalar. In a higher-rank tensor, all uncontracted indices remain free. [a]

[a] Lecture timestamp: 01:09:00.

For example,

$$A^{mn}{}_{pq} \longmapsto A^{mn}{}_{pn} = B^m{}_p \tag{1.24}$$

contracts n and lowers the rank by two. The precise letters are dummy labels; the structural rule is that one upper and one lower slot are summed.

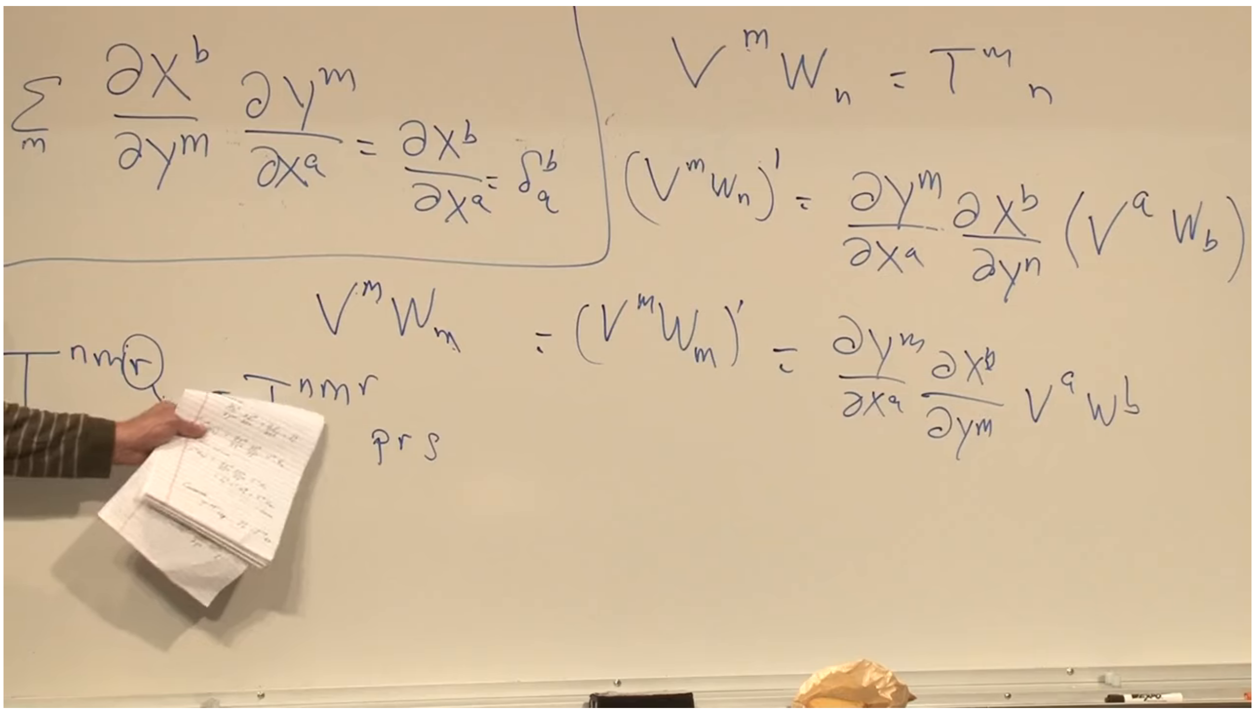

Figure 1.22: A higher-rank contraction removes one upper and one lower index while preserving the remaining free indices.

Lecture frame: 01:12:35.

Question from the audience. Why not contract two upper indices, or two lower indices, in the same way?

Response. Two upper indices supply two Jacobians of the same orientation, $\partial y^m/\partial x^a$ and $\partial y^m/\partial x^b$. They do not multiply to the identity, so the proposed sum does not have a tensor transformation law. A metric can first lower one of the indices, after which an upper–lower contraction is legal; without that additional tensor, the operation is not coordinate invariant. [a]

[a] Lecture timestamp: 01:13:03.

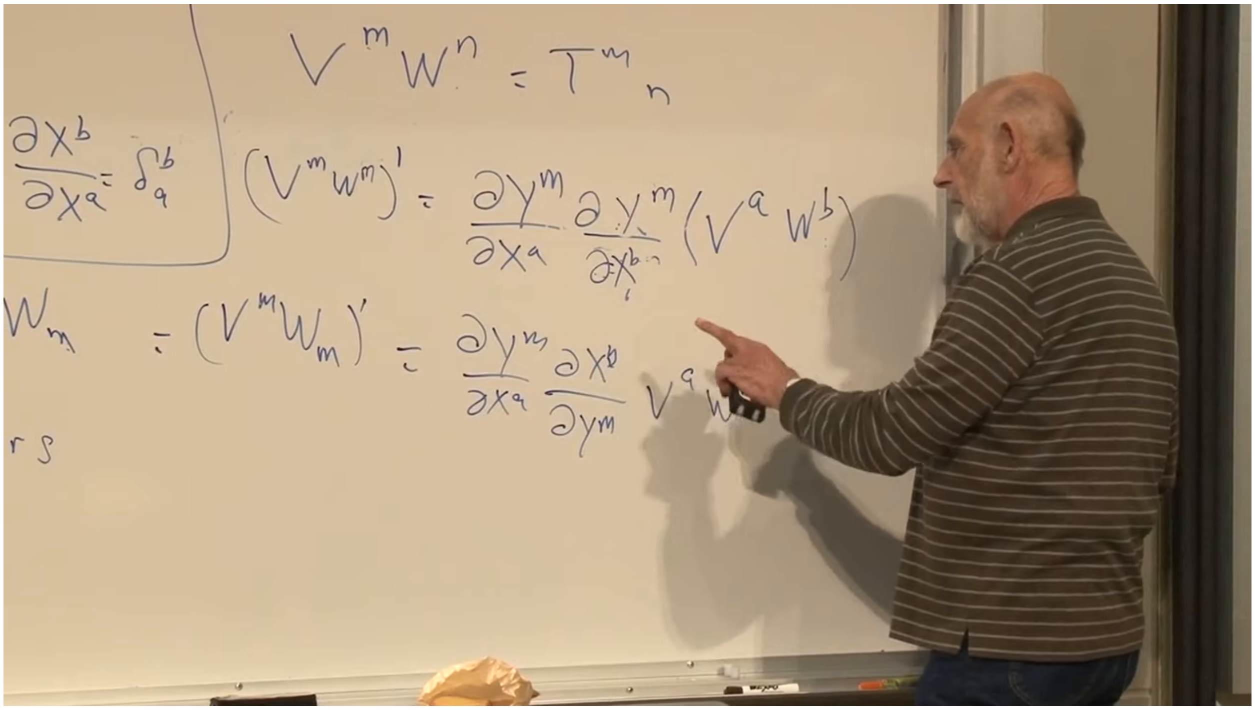

Figure 1.23: The attempted same-variance contraction leaves two same-direction Jacobians and therefore does not collapse to an identity.

Lecture frame: 01:14:30.

Question from the audience. Is contraction the tensor version of an inner product?

Response. A contraction pairs a vector slot with a covector slot and is the basic invariant pairing. With a metric, two vectors can be turned into such a pair, so the familiar inner product is a metric followed by contraction. [a]

[a]Lecture timestamp: 01:14:43.

Question from the audience. Must the two contracted indices run over the same range?

Response. Yes. They refer to dual bases of the same d-dimensional tangent space and cotangent space. Otherwise the summation and the inverse-Jacobian identity would not be defined. [a]

[a]Lecture timestamp: 01:15:13.

Question from the audience. Why does a tensor that vanishes in one frame vanish in every frame?

Response. Every transformed component is a linear combination of the original components. All those combinations vanish when every input component is zero. This simple linearity is what makes tensor equations coordinate independent. [a]

[a]Lecture timestamp: 01:15:31.

1.8 The Metric from Squared Distance

Return now to geometry. For an infinitesimal displacement with components dx^m, define its squared length by

$$ds^2 = g_{mn}(x)\, dx^m dx^n. \tag{1.25}$$

This is the generalized Pythagorean theorem. The metric may vary from point to point, and off-diagonal terms record nonorthogonality of the coordinate directions.

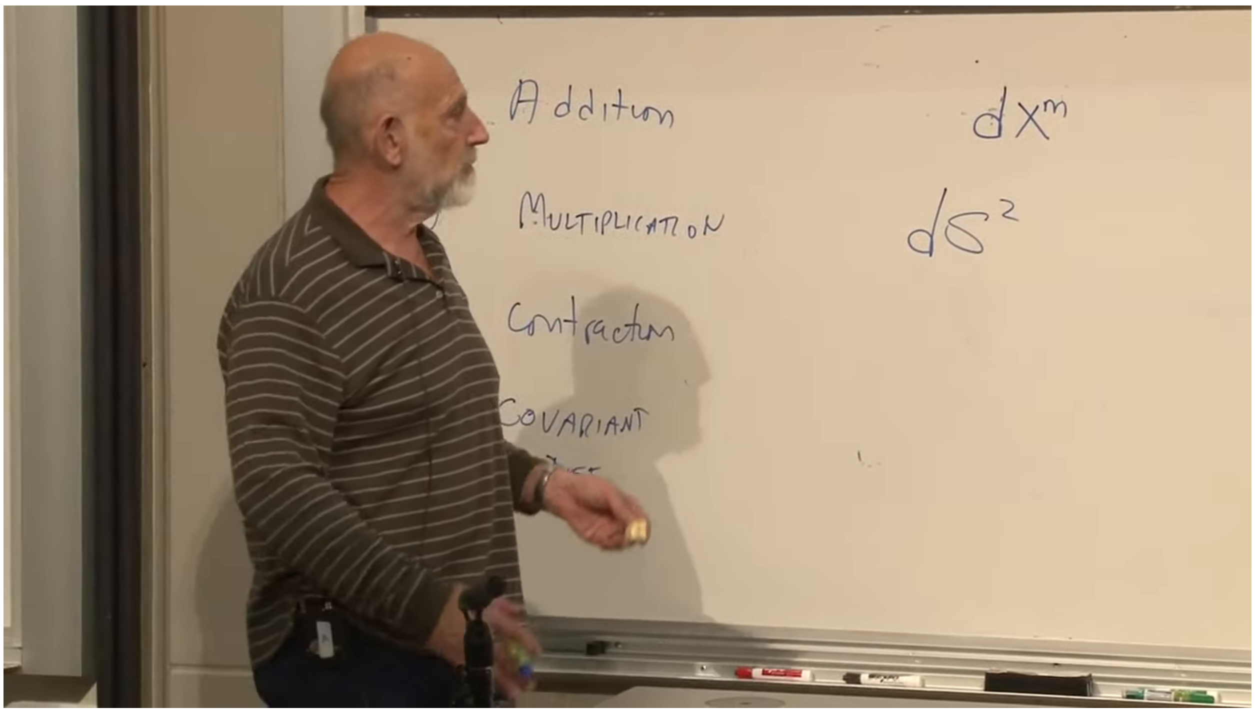

Figure 1.24: The tensor-operation list gives way to the quadratic definition of the infinitesimal line element.

Lecture frame: 01:18:45.

Only the symmetric part contributes because $dx^m dx^n = dx^n dx^m$. We may therefore take

$$g_{mn} = g_{nm}. \tag{1.26}$$

In d dimensions a symmetric metric has

$$\frac{d(d+1)}{2} \tag{1.27}$$

independent components: ten in four dimensions, six in three, and three in two.

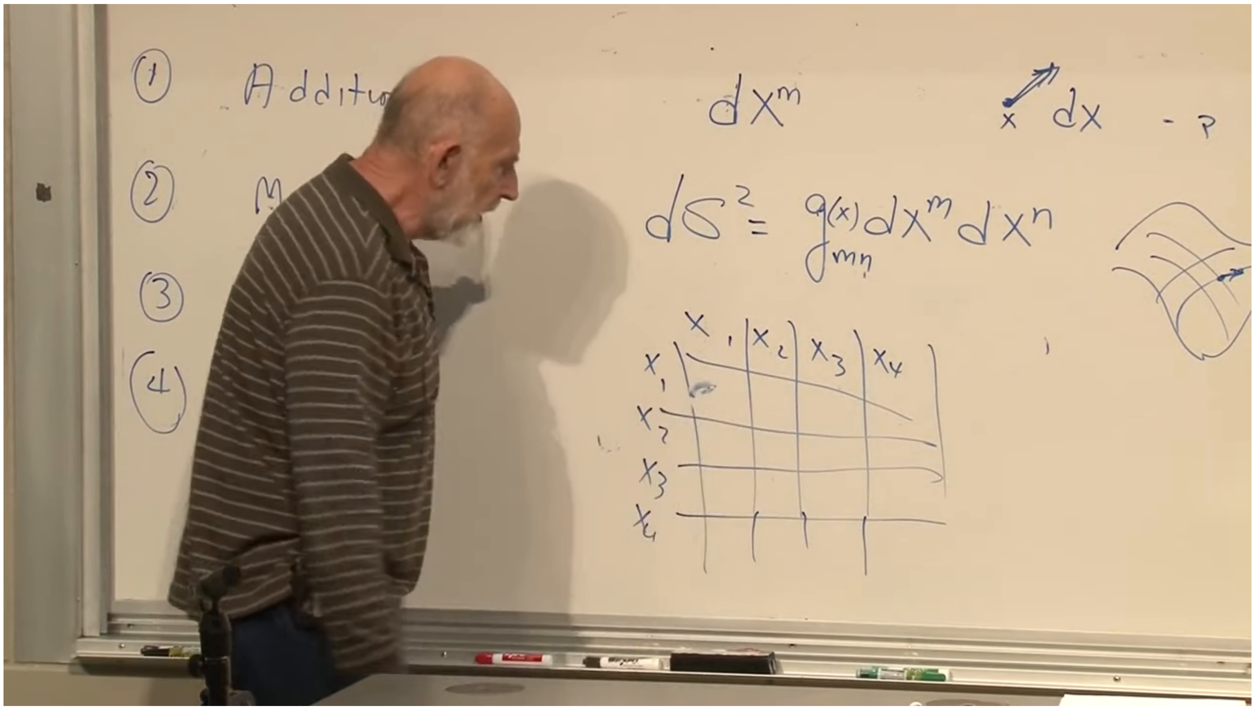

Figure 1.25: The line element and the symmetric component array used to count the independent entries of the metric.

Lecture frame: 01:21:35.

Question from the audience. Why are there ten independent metric components in four dimensions rather than sixteen?

Response. There are four diagonal entries and six unordered off-diagonal pairs. Equivalently, $4(4+1)/2 = 10$. The antisymmetric part would multiply the symmetric product $dx^m dx^n$ and cancel out. [a]

[a]Lecture timestamp: 01:19:50.

Question from the audience. How many independent components are there in three dimensions?

Response. Six: three diagonal entries and three off-diagonal pairs, or $3(3+1)/2$. [a]

[a]Lecture timestamp: 01:20:47.

Question from the audience. And in two dimensions?

Response. Three: g_{11}, g_{22}, and the single symmetric pair $g_{12} = g_{21}$. [a]

[a]Lecture timestamp: 01:22:28.

1.8.1 Why the metric is a tensor

Squared distance is a scalar. Writing the same displacement in two coordinate systems therefore gives

$$g_{mn}(x)\, dx^m dx^n = g'_{pq}(y)\, dy^p dy^q. \tag{1.28}$$

Use

$$dx^m = \frac{\partial x^m}{\partial y^p} dy^p \tag{1.29}$$

twice:

$$ds^2 = g_{mn}(x) \frac{\partial x^m}{\partial y^p} \frac{\partial x^n}{\partial y^q} dy^p dy^q. \tag{1.30}$$

Comparison with the primed expression identifies

$$g'_{pq}(y) = \frac{\partial x^m}{\partial y^p} \frac{\partial x^n}{\partial y^q} g_{mn}(x). \tag{1.31}$$

This is exactly the transformation law for a covariant rank-two tensor.

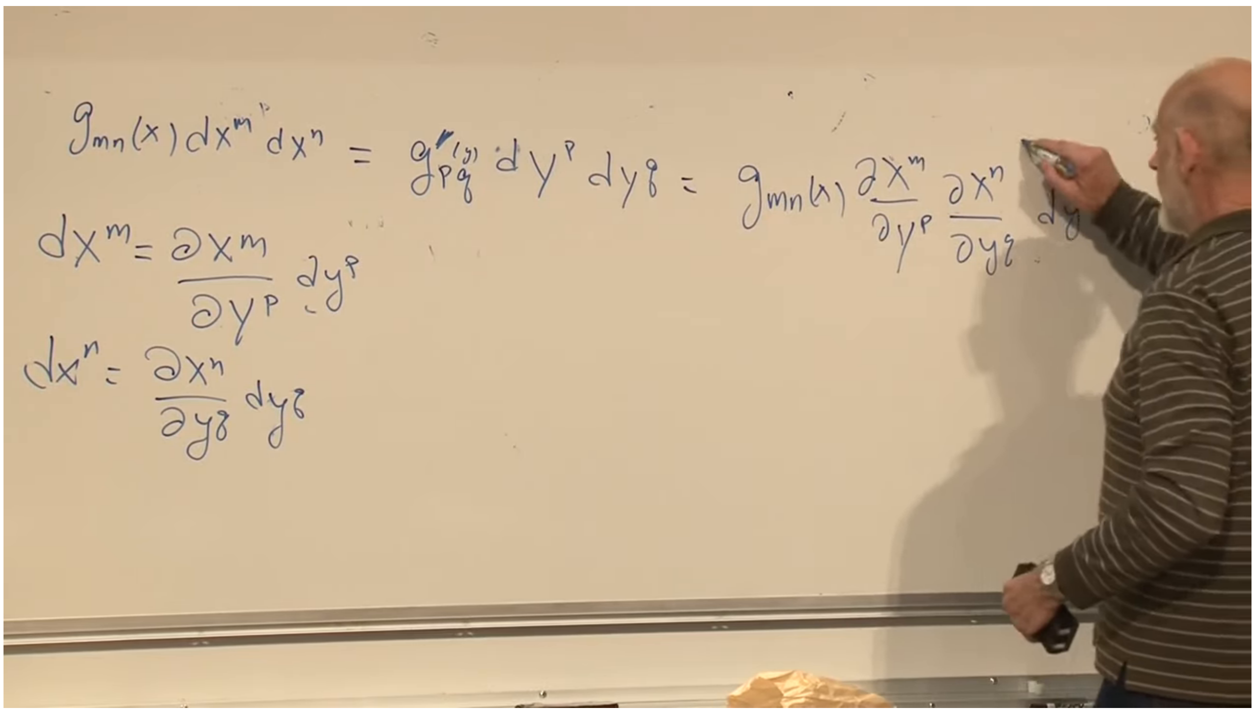

Figure 1.26: Derivation of the metric transformation law from invariance of the scalar line element. *Lecture frame: 01:26:25.*

The conclusion also explains the two lower indices. Each lower metric index pairs with one upper index carried by a displacement component. Under a coordinate change, the two inverse Jacobians in (1.31) cancel the two forward Jacobians of the differentials, leaving ds^2 unchanged.

1.9 The Metric Matrix and Its Inverse

At a point, g_{mn} is a symmetric matrix:

$$[g_{mn}] = \begin{pmatrix} g_{11} & g_{12} & \cdots \\ g_{12} & g_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \tag{1.32}$$

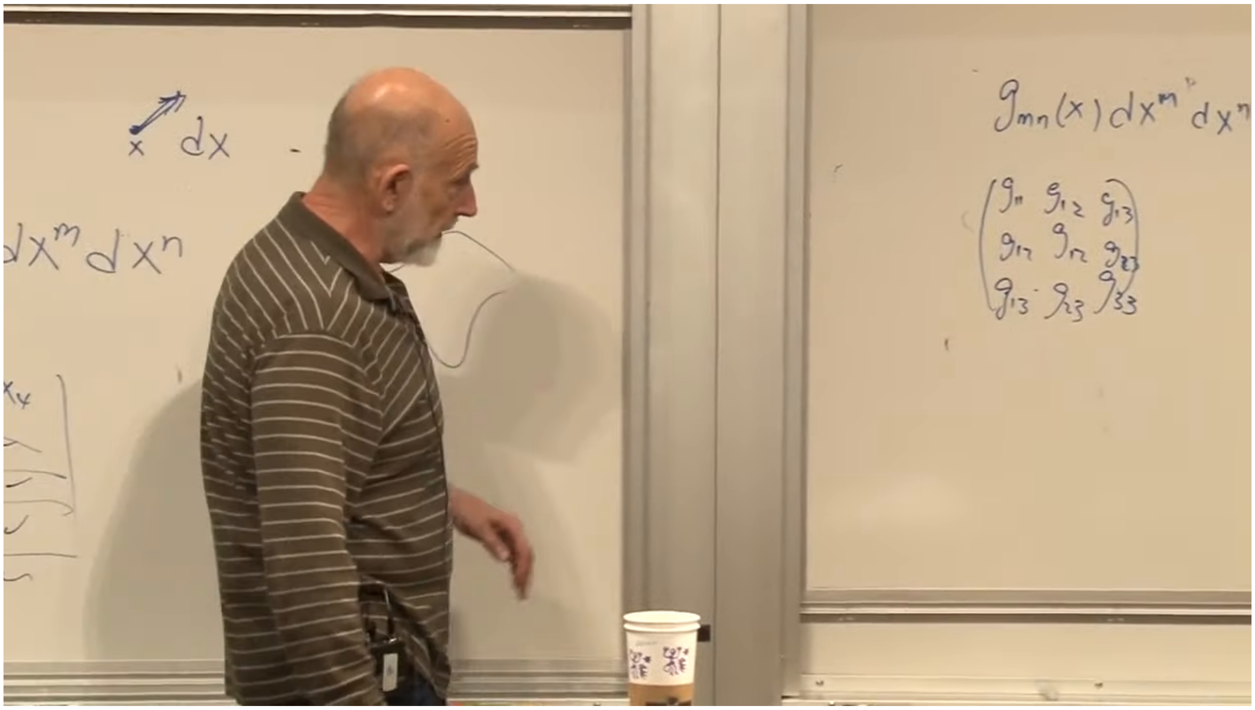

Figure 1.27: The metric displayed as a symmetric matrix after its tensor transformation law has been established.

Lecture frame: 01:28:50.

The metric is required to be nondegenerate: it has no zero eigenvalue and therefore possesses an inverse. Denote that inverse by g^{mn}. In index notation,

$$g_{mn}g^{np} = \delta_m{}^p. \tag{1.33}$$

The repeated n is summed, while m and p label the row and column of the product.

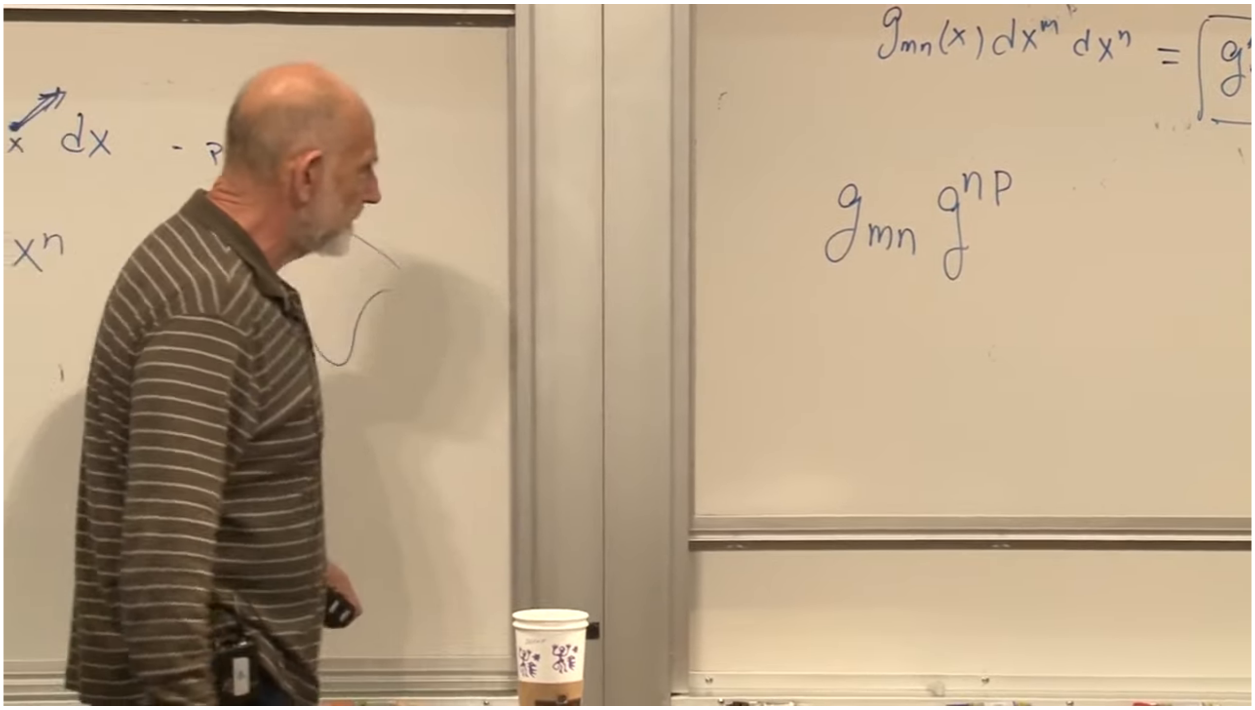

Figure 1.28: The inverse metric introduced through $g_{mn}g^{np} = \delta_m{}^p$.
Lecture frame: 01:31:40.

Question from the audience. Where do the vectors and tensors at a point actually live?

Response. Vectors at a point form its tangent space $T_P M$; covectors form the dual space $T_P^* M$. Higher tensors belong to tensor products of these spaces. A tensor field assigns such an object smoothly to every point. The metric is a bilinear form on the tangent space, and its inverse is the corresponding bilinear form on the cotangent space. [a]

[a]Lecture timestamp: 01:34:15.

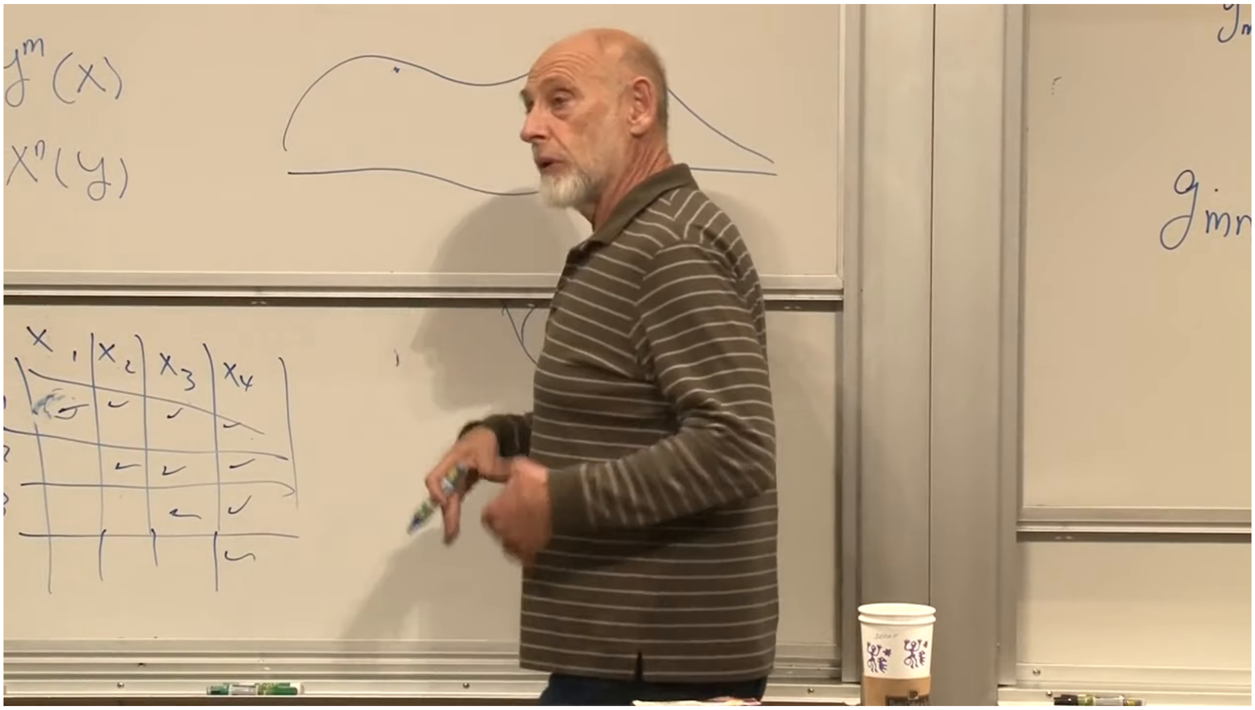

Figure 1.29: A vector drawn in the tangent space attached to a point of a curved manifold.
Lecture frame: 01:35:45.

Question from the audience. Does $g_{mn}g^{np} = \delta_m{}^p$ refer only to the case $m = p$?

Response. No. It is the complete matrix equation. For every ordered pair (m, p), the summed product is one when $m = p$ and zero when $m \neq p$. All diagonal and off-diagonal equations are present simultaneously. [a]

[a] Lecture timestamp: 01:36:50.

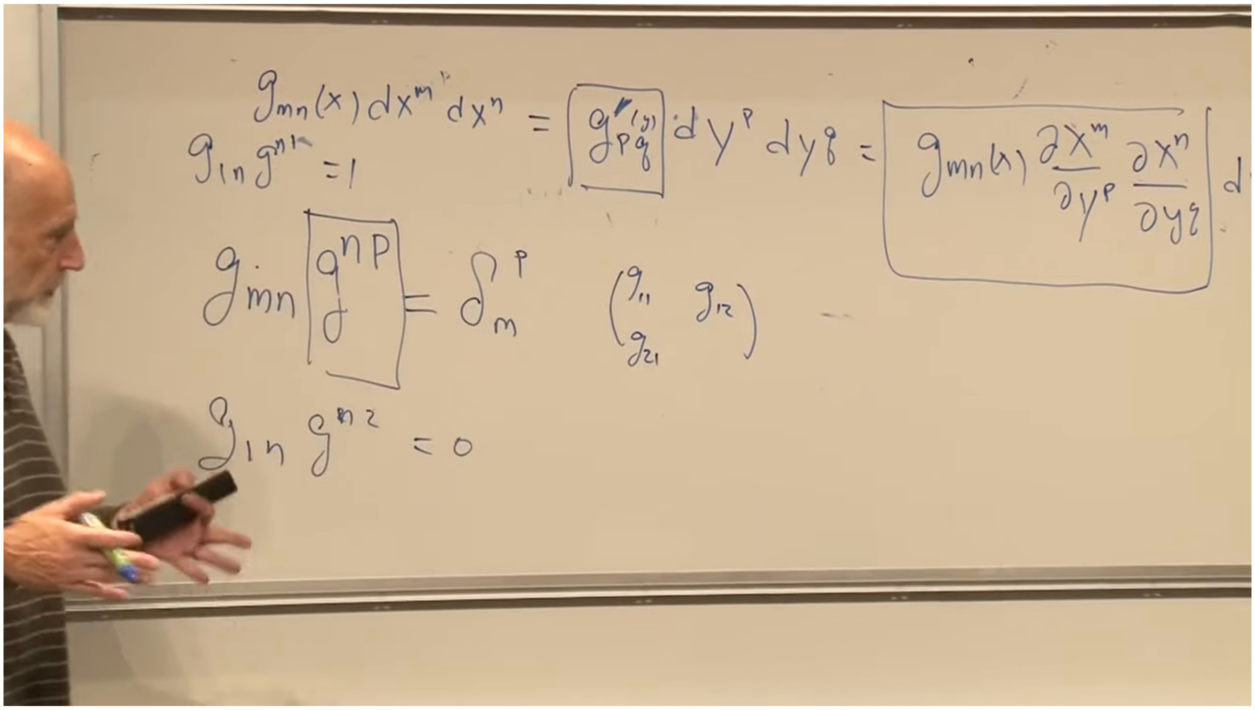

Figure 1.30: The inverse relation interpreted as the complete mixed-index identity matrix.
Lecture frame: 01:38:50.

In two dimensions this means

$$\begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix} \begin{pmatrix} g^{11} & g^{12} \\ g^{12} & g^{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1.34}$$

For example,

$$g_{11}g^{11} + g_{12}g^{12} = 1, \tag{1.35}$$
$$g_{11}g^{12} + g_{12}g^{22} = 0. \tag{1.36}$$

The other row supplies the remaining two equations.

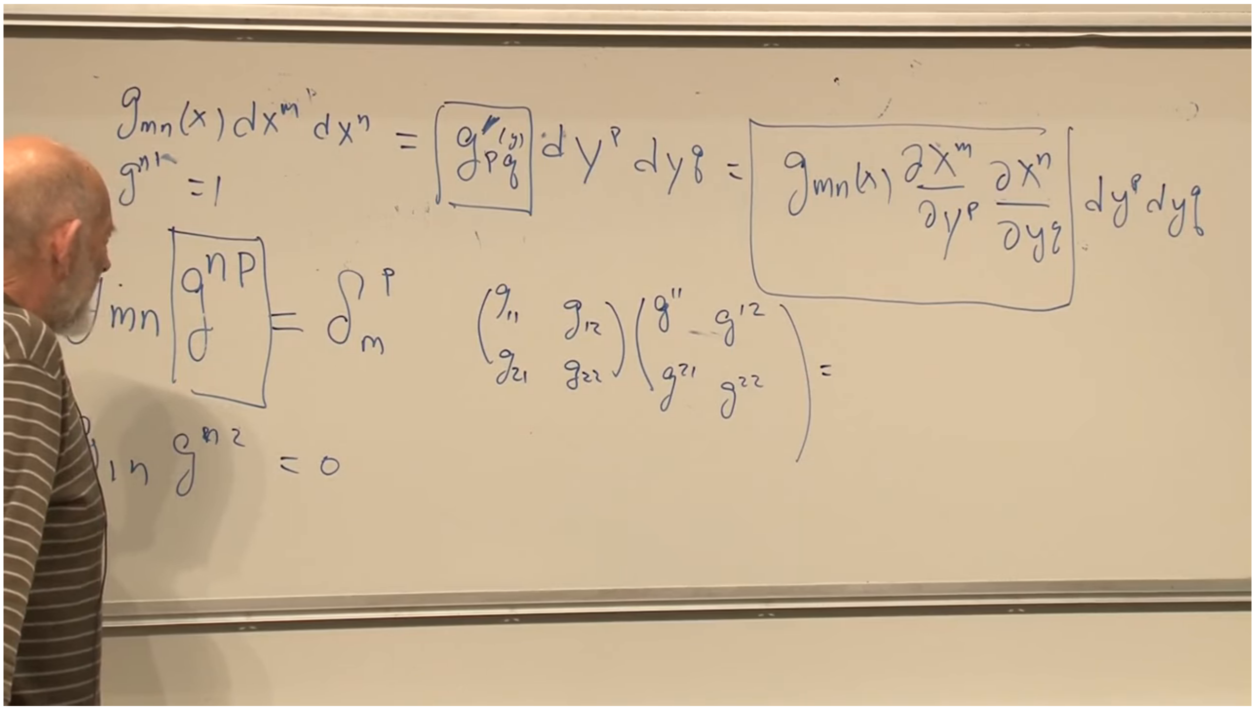

Figure 1.31: The inverse-metric identity expanded as an explicit two-dimensional matrix product.

Lecture frame: 01:40:00.

Question from the audience. Must the metric be positive definite?

Response. For ordinary Riemannian geometry, positive definiteness makes every nonzero squared length positive. Spacetime uses a Lorentzian metric instead, with one sign opposite to the other three. What both cases require here is nondegeneracy, not positivity. [a]

[a]Lecture timestamp: 01:42:18.

Question from the audience. Why exclude a zero eigenvalue?

Response. A zero eigenvalue makes the metric matrix singular, so g^{mn} does not exist and the metric cannot establish a one-to-one correspondence between vectors and covectors. A metric, in the usual differential-geometric sense, is required to be nondegenerate. [a]

[a]Lecture timestamp: 01:43:56.

Question from the audience. Do the ten independent components of a four-dimensional metric have anything to do with ten-dimensional string theory?

Response. No. The first ten is simply $4(4+1)/2$, the number of entries in a symmetric four-by-four matrix. The appearance of ten spacetime dimensions in superstring theory has a different origin. [a]

[a]Lecture timestamp: 01:44:32.

Question from the audience. If Lorentzian spacetime has nonzero vectors of zero length, does that make the metric noninvertible?

Response. No. A null vector satisfies $g_{mn}V^mV^n = 0$, a cancellation within an indefinite quadratic form. It is not a zero-eigenvalue vector satisfying $g_{mn}V^n = 0$. For example, the Minkowski metric has eigenvalues ± 1 and is perfectly invertible while still admitting lightlike vectors. [a]

[a] Lecture timestamp: 01:45:02.

1.10 What Comes Next

We can now state the geometric program cleanly. The metric records infinitesimal distance, transforms covariantly, and has an inverse. Its components may look complicated merely because the coordinates are complicated. To decide whether the geometry itself is curved, we must compare the metric at neighboring points in a coordinate-independent way.

Ordinary derivatives are not enough, because differentiating a tensor also differentiates the position-dependent Jacobians in its transformation law. The next step is therefore covariant differentiation. From derivatives of the metric one constructs a connection; from derivatives and products of that connection one constructs curvature. Only then will the curled-page question receive its final intrinsic answer.